

Electrical and Magnetic Properties of Materials

1. INTRODUCTION

The electrons in the outermost shell of atoms controls the electrical and magnetic behaviour of substances. Recent developments in the field of electronic structure of substances have accelerated the growth of number of useful solid state devices. The prime objective of this chapter is to explore the electrical and magnetic properties of materials.

2. ELECTRICAL CONDUCTION

One of the most important characteristics of a solid material is the ease with which it transmits an electric current. Ohm's law relates the current I (i.e., the time rate of charge passage, Q/t , $Q \rightarrow$ Charge and $t \rightarrow$ time) to the applied voltage V as follows:

$$V = I R \quad (1)$$

where R is the resistance of the material through which the current I is passing. The units of V , I and R are Volt, ampere and Ohm respectively. The electrical resistivity (ρ) of solids is probably the most important of all physical properties. The value of electrical resistance is influenced by specimen configuration, and for many materials is independent of current. However, the resistivity (ρ) of material is independent of specimen geometry but it is related to R through the relation

$$\rho = \frac{RA}{l} \quad (2)$$

where l is the distance between the two points of the specimen at which the voltage is measured, and A is the area of cross-section perpendicular to the direction of the current. The units of ρ are Ohm-meters ($\Omega - m$). From Eqs. (1) and (2),

$$\rho = \frac{VA}{Il} \quad (3)$$

Many factors influence the value of ρ for a given material. Values of resistivity of common materials at 20°C are given in Table 14.1 and few engineering materials are given in Table 14.1(a).

Table 14.1 Resistivity of some common materials at 20°C

Material	Resistivity (ρ) ($\Omega - m$)	Material	Resistivity (ρ) ($\Omega - m$)
Silver	1.6×10^{-8}	Germanium	10^{-5} to 0.6
Aluminium	2.66×10^{-8}	Silicon	10^{-5} to 2.5×10^3

(Contd.)

Table 14.1 (Contd.)

Material	Resistivity (ρ) ($\Omega - m$)	Material	Resistivity (ρ) ($\Omega - m$)
Iron	9.1×10^{-8}	PVC	1.0×10^{10}
Copper	1.67×10^{-8}	Bakelite	1.0×10^{11}
Nickel	13.3×10^{-8}	Mica	1.0×10^{11}
Carbon-steel	1.7×10^{-7}	Glass	1.0×10^{16}
Polythene	1.0×10^{16}	Stainless steel	7.0×10^{-7}
Graphite	1.4×10^{-7}	Steatite Porcelain	1.0×10^{13}
		Alumina	10^{11}
		Diamond	10^{12}

Table 14.1(a) Room temperature electrical resistivity for few engineering materials

Material	Electrical Resistivity ($\Omega - m$)
Steel alloy 1020 (annealed)	1.60×10^{-7}
Steel alloy 4140 (quenched and tempered)	2.20×10^{-7}
Steel alloy 4340 (quenched and tempered)	2.48×10^{-7}
Stainless steel alloy 440 A (annealed)	6.0×10^{-7}
Stainless steel alloy 17-7 PH (annealed)	8.3×10^{-7}
Gray irons	
• Grade G 1800	15.0×10^{-7}
• Grade G 3000	9.5×10^{-7}
• Grade G 4000	8.5×10^{-7}
Ductile irons	
• Grade 60-40-18	5.5×10^{-7}
• Grade 80-55-06	6.2×10^{-7}
• Grade 120-90-02	6.2×10^{-7}
Aluminium alloy 7075 (T 6 treatment)	5.22×10^{-8}
Aluminium alloy 356.0 (T 6 treatment)	4.42×10^{-8}
Copper alloy C 3600 (free cutting brass)	6.6×10^{-8}
Copper alloy C 71500 (copper-nickel, 30%)	37.5×10^{-8}
Magnesium alloy A Z 91 D	17.0×10^{-8}
Titanium alloy Ti - 5Al - 2.5Sn	15.7×10^{-7}
Nickel 200	0.95×10^{-7}
Inconel 625	12.90×10^{-7}
Monel 400	5.47×10^{-7}
Haynes alloy 25	8.9×10^{-7}
Invar	8.2×10^{-7}
Super invar	8.0×10^{-7}
Kovar	4.9×10^{-7}
Lead-tin-solder (60 Sn - 40 Pb)	1.50×10^{-7}

The resistivity of some widely used metal and their alloys along with their temperature coefficient are given in Table 14.2.

Table 14.2 Electrical resistivity of some metals and their alloys

Metals and alloys	Resistivity (ρ) at 20°C ($\Omega\text{-m}$)	Temperature coefficient $\alpha^\circ\text{C}$
• Copper, annealed	1.67×10^{-8}	4.29×10^{-3} (0-100°C)
• Copper, reduced 75% by cold drawing	1.71×10^{-8}	—

(Contd.)

Table 14.2 (Contd.)

Metals and alloys	Resistivity (ρ) at 20°C ($\Omega\text{-m}$)	Temperature coefficient $\alpha^\circ\text{C}$
• Cartridge brass annealed 70% Cu and 30% Zn	6.20×10^{-8}	1.48×10^{-3} (20°C)
• Aluminium, annealed	2.65×10^{-8}	4.29×10^{-3} (20°C)
• Iron, annealed	9.71×10^{-8}	6.57×10^{-3} (20°C)
• Constantan 55% Cu and 45% Ni	49×10^{-5}	0.02×10^{-3} (25°C)
• Manganin 84% Cu, 12% Mn and 4% Ni	44×10^{-7}	0.009×10^{-3} (25°C)
		-0.42×10^{-3} (100°C)
• Nichrome 80% Ni and 20% Cr	108×10^{-5}	0.14×10^{-3} (0-500°C)

3. ELECTRICAL CONDUCTIVITY (σ)

Sometimes, electrical conductivity (σ) is used to specify the electrical character of a material. Electrical conductivity is simply the reciprocal of resistivity (ρ), i.e.

$$\sigma = \frac{1}{\rho} \quad (4)$$

Electrical conductivity is indicative of the ease with which a material is capable of conducting an electrical current. The units of σ are reciprocal Ohm-meters [$(\Omega\text{-m})^{-1}$] or mho/m. It is also expressed in Siemens/m.

When an electric field E is applied to a conductor an electric current begins to flow and the current density by Ohm's law is

$$J = \sigma E \quad (5)$$

The conductivity may be defined as the movement of electrical charge from one point to another and it depends on the number of charge carriers (n), the charge per carrier (e) and the mobility of carriers (μ), i.e.

$$\sigma = ne\mu \quad (6)$$

The unit of mobility (μ) is $\text{m}^2 \text{V}^{-1} \text{s}^{-1}$. Mobility is an important term in the study of semiconductors.

One can obtain the expression (6) from first principle as follows:

Let E be the electric field applied to a conductor, e be the charge on the electron and m the mass of the electron. The electrons move in a specific direction under the influence of the electric field. The directional motion of a free electron is called a *drift*. The average velocity gained during this drift motion is termed as *drift velocity*.

As electron is being negatively charged particle, the force acting on it under the electric field intensity E is,

$$F = -eE \quad (7)$$

The electron drift is in a direction opposite to that of the applied field. During the accelerated motion, the electron collides with the defects in the lattice. As a result of the consequence scattering, the electron loses the velocity it gained from the electric field. The effect of the crystal lattice may be reduced considerably due to a retarding force (may be due to damping). This force is proportional to the velocity v and mass m of the electron. The retarding force is represented as $-\alpha mv$, where α is a constant. We can write the equation of motion of the electron as

$$m \frac{dv}{dt} = -eE = -\alpha mv \quad (8)$$

or

$$dv = -\frac{eE}{m} dt$$

or

$$v = -\frac{eE}{m} t + \text{constant} \quad (9)$$

If the average time between collisions is 2τ then, during this time, the electron is acted upon by a force F given by Eq. (7). But at $t = 0$, $v = 0$ (immediately after each collision) and hence the integration constant in (9) is zero. Thus (9) takes the form

$$v = -\frac{eE}{m} \tau$$

Obviously, the mean velocity $= -eE \tau/m$, where τ is called the relaxation time and is time interval in which there is unit probability of a collision.

When there is a collision of lattices (resistance), the current density J due to n electrons per unit volume of charge e and drift velocity v is expressed as

$$J = nev \quad (10)$$

$$= ne \left(\frac{eE\tau}{m} \right) \quad (11)$$

$$= \sigma E \quad (12)$$

$$\therefore \sigma = \frac{J}{E} = \frac{ne^2 \tau}{m} = \frac{ne(e\tau)}{m} \quad (13)$$

The velocity in a unit electric field, i.e. v/E is the mobility (μ) of the electron. Hence

$$\mu = \frac{v}{E} = \frac{1}{E/v} \left(\frac{Ee\tau}{m} \right) = \frac{e\tau}{m}$$

Using Eqs. (10) and (12), one obtains

$$\begin{aligned} J &= ne\mu E \\ \sigma &= ne\mu \end{aligned} \quad (14)$$

Equation (14) is of immense importance for solid materials. From (14), we note that the electrical conductivity depends on two factors: (i) the number n of charge carriers per unit volume and (ii) their mobility, μ .

Solid materials exhibit an amazing range of electrical conductivities, extending over 27 orders of magnitude; probably no other physical property experiences this breadth of variation. In fact, one way of classifying solid materials is according to the ease with which they conduct an electric current. Within this classification scheme, one can classify materials into three broad categories:

(i) *Conductors* (ii) *Semi conductors* and (iii) *Insulators*

Metals and their alloys are good conductors. Copper, silver and gold are among the best conductors of electricity, followed by aluminium, iron and nickel. Some semimetals, e.g. graphite also fall in this group. According to free electron theory, when the outer orbit of an atom has less than one half of the maximum 8 electrons, the material is usually a metal and a good conductor of electricity. Metals have conductivities on the order of $10^7 (\Omega\text{-m})^{-1}$. The electrical resistivity of conductor range from 10^{-9} to $10^{-4} (\Omega\text{-m})$.

Semiconductors are materials which behave are insulators at 0 °K but a significant rise in electrical conductivity is observed as the temperature rises. At room temperature, the electrical conductivity of semiconductor falls between that of a conductor and insulator. The electrical conductivity semiconductors ranges from about 10^5 to $10^{-7} (\Omega\text{-m})^{-1}$, as compared with the conductivity on the order of $10^7 (\Omega\text{-m})^{-1}$ for good conductors and minimum conductivity of $10^{-15} (\Omega\text{-m})^{-1}$ for good insulators. Semiconductors form the base materials for a number of electronic devices. Germanium and silicon are the widely used common semiconductors. According to free electron theory, when the outer orbit of an atom has exactly one-half the maximum eight electrons, the material has both metal and non-metal properties and usually exhibit semiconducting properties. We must note that the electrical resistivity of a semiconductor is usually strongly dependent on temperature.

There are also materials with very low conductivities, ranging between 10^{-10} and $10^{-20} (\Omega\text{-m})^{-1}$. These are electrical *insulators*. Mica, PVC, rubber, porcelain and bakelite are few examples of insulators. The

resistivity range of insulators extends from 10^4 to 10^{17} ($\Omega\text{-m}$). According to electron theory, when the outer orbit of an atom has more than one half of the maximum eight electrons, the material is usually a non-metal and non-conductor.

A brief comparison between metals, semiconductors and insulators is presented in Table 14.3.

Table 14.3 A comparison between conductors, semiconductors and insulators

<i>Conductors</i>	<i>Semiconductors</i>	<i>Insulators</i>
1. Conductivity decreases with increase in temperature upto nearly zero value	• Conductivity increases with increase in temperature. Conductivity is particularly sensitive to impurity type and content.	• Conductivities increases with increase in temperature
2. Conductivity of metals is of the order of 10^7 ($\Omega\text{-m}$) ⁻¹	• Conductivity of semiconductors range from 10^{-6} to 10^4 ($\Omega\text{-m}$) ⁻¹	• Conductivity of insulators range between 10^{-10} to 10^{-20} ($\Omega\text{-m}$) ⁻¹
3. Electrical resistivity is very low and range from 10^{-9} to 10^{-4} ($\Omega\text{-m}$)	• Resistivity is normally high and range between 10^{-3} to 10^3 ($\Omega\text{-m}$)	• Resistivity is very high and range between 10^4 to 10^{17} ($\Omega\text{-m}$)
4. Temperature coefficient of resistance is not constant	• Temperature coefficient of resistance is negative	• Negative resistance temperature coefficient. Probably with the rise in temperature some electrons reach to the conduction band.
5. At low temperatures they exhibit semiconductivity. At very low temperatures, some elements and their alloys exhibit infinite conductivity, i.e. super conductivity	• At low temperature semiconductors become <i>dielectrics</i> (insulators)	• No change in the properties of the insulators observed
6. Have unfilled overlapping energy bands	• Have filled energy bands and small forbidden zones	• There is a large energy gap in between valence and conduction band
7. Current carriers in conductors are free electrons which exist whether external field is applied or not	• Current carriers are originated due to absorption of electrical, radiant or thermal energy from external source. Electrons and holes, both serve as current carriers	• Energy required for electrons to cross the energy gap between conduction band and valence band is very large and hence no conduction

4. ELECTRONIC AND IONIC CONDUCTION

We have seen that an electric current results from the motion of electrically charged particles in response to the forces that act on them from an externally applied field. Positively charged particles are accelerated in the electric field direction, and negatively charged particles in the direction opposite to the electric field direction. A current arises within most solid materials due to flow of electrons and this is termed as *electronic conduction*. In addition, for *ionic* materials a net motion of charged ions is possible that produces a electric current; such is termed *ionic conduction*.

5. BAND STRUCTURE IN SOLIDS

A solid may be thought of as consisting of a large number, say N , of atoms initially separated from one another, which are subsequently brought together and bonded to form the ordered atomic arrangement exhibited by the crystalline material. At relatively large separation distances, each atom is independent of all the other atoms as will have the atomic energy levels and electron configuration as if isolated. However, as the atoms in a solid come within the close proximity of one another, electrons are acted upon, or

perturbed, by the electrons and nuclei of adjacent atoms. This influence is such that each distinct atomic state may split into a series of closely spaced electron states in the solid, to form what is termed an *electron energy band*. The extent of splitting of atomic states depends on interatomic separation (Fig. 14.1) and begins with the outermost electron shells, since they are the first to be perturbed as the atoms coalesce. Within each band, the energy states are discrete, yet the difference between adjacent states is exceedingly small. At the equilibrium spacing, band formation may not occur for the electron subshells nearest the nucleus (Fig. 14.2(b)). Moreover, gaps may exist between adjacent bands, as also shown in Fig. 14.2(a), normally energy lying within these band gaps are not available for electron occupancy. The conventional way of representing electron band structure is shown in Fig. 14.2(a).

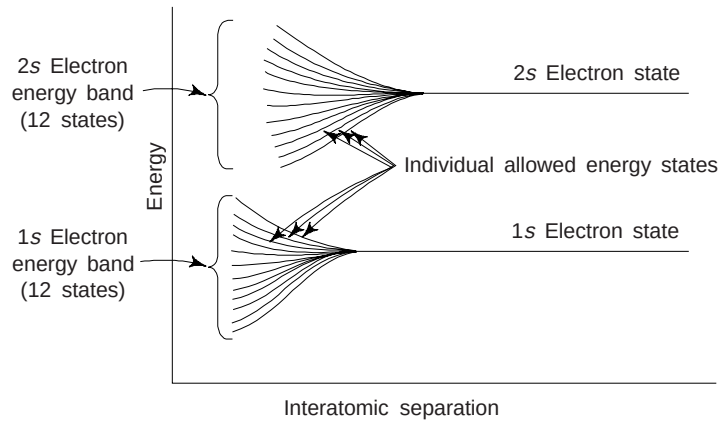


Fig. 14.1 A schematic plot of electron energy vs. interatomic separation of an aggregate of 12 atoms, i.e. for $N = 12$. Upon close approach, each of the $1s$ and $2s$ atomic states splits to form an electron energy band consisting of 12 states

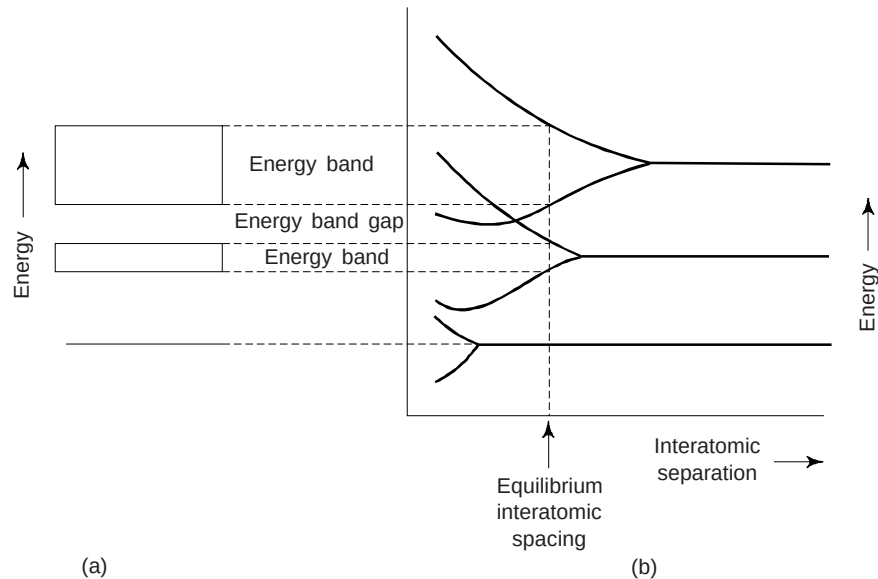


Fig. 14.2 (a) The conventional representation of the electron energy band structure for a solid material at the equilibrium interatomic separation. (b) Electron energy vs. interatomic separation for an aggregate of atoms, exhibiting how the energy band structure at the equilibrium separation in (a) is generated

The number of states within each band will equal the total of all states contributed by the N atoms, e.g. a s -band will consist of N -states, and a p -band of $3N$ states. With regard to occupancy, each energy state will accommodate two electrons, which must have oppositely directed spins. Moreover, bands will contain the electrons that resided in the corresponding levels of the isolated atoms; for e.g., a $4s$ energy band in the solid will contain those isolated atom's $4s$ electrons. Of course, there will be empty bands and, possibly, bands that are only partially filled.

The electrical properties of a solid material are a consequence of its *electron band structure*, i.e. the arrangement of outermost electrons in bands and the way in which they are filled with electrons. A distinctive band structure type exists for metals, for semiconductor, and for insulators.

The band structure in a solid determines whether the solid is an insulator or a conductor or a semiconductor. The bands are filled up to a certain level by the electrons within each atom. The highest band in which electrons are still predominantly attached to their atoms are found is called *valence band*. This is the band in which the valence (outermost) electrons from each atom will be located. These are the electrons that are the possible carriers of electricity. However, in order for an electron to conduct, it must get up to slightly higher energy so that it is free of the grip of its atom. At 0 K four different types of band structures are possible. In the first (Fig. 14.3(a)), one outermost band is only partially filled with electrons. The energy corresponding to the highest filled energy state of 0 K is called the Fermi energy E_F , as indicated in future. This energy band structure is typified by some metallic conductors in particular those that have a single s valence electron, e.g. copper, etc. Each copper atom has one $4s$ electron. However, for a solid comprised of N atoms, the $4s$ band is capable of accommodating $2N$ electrons. Obviously, only half the available electron positions within this $4s$ band are filled.

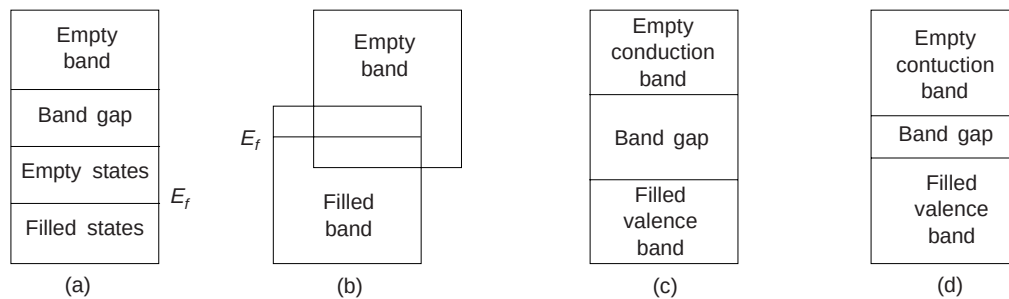


Fig. 14.3 Various possible electron band structure in solids at 0 K. (a) and (b) for conductors. Band structure (a) is found in metals such as copper, in which there are available electron states above and adjacent to filled states, in the same band. Band structure (b) is for metals such as magnesium, where there is an overlap of filled and empty outer bands. (c) The electronic band structure characteristic of insulators; the band gap > 2 eV or more. (d) The electron band structure found in semiconductors; $E_g < 2$ eV

For the second band structure, also found in metallic conductors (Fig. 14.3(b)), there is an overlap of an empty band and a filled band. Magnesium has this type of band structure. Each isolated Mg has two $3s$ electrons. However, when a solid is formed, the $3s$ and $3p$ bands overlap. In this instance and at 0K, the Fermi energy, E_F is taken as that energy below which, for N atoms, N states are filled, two electrons per state. Therefore, the band theory tells us that we have a conductor, when

- (i) the *valence band* is not filled, so electron can move to higher states in the valence band and be free (energy gaps between valence band and conduction band are very small), or
- (ii) when there is no energy gap between the valence band and the conduction band, i.e. there is overlapping, so electrons can easily make the transitions from the valence to the conduction bands.

The final two band structures (Fig. 14.3(c) and (d)) are similar; one band (the *valence band*) that is completely filled with electrons that is separated from an empty *conduction band* and an *energy band gap* lies between them. When energy band gap is of the order of 1 eV (for Si, $E_g = 1.12$ eV; and for Ge, $E_g = 0.72$ eV). These materials with narrow band gap are called *semiconductors*. These are discussed in chapter 15.

When the energy band gap is relatively wide ~ 5 eV or even more, the materials are called insulators. The band theory of solids tells us that an insulator is a material in which the valence band (VB) are filled and the energy band gap E_g between VB and conduction band (CB) is too large so that valence electrons cannot jump at normal temperatures from VB to CB. An insulator does not conduct at 0 K or even at room temperature because there are no conduction electrons in it. However, an insulator may conduct if its temperature is very high or if a high voltage is applied across it. This is known as *breakdown* of an insulator.

6. CONDUCTION IN TERMS OF BAND AND ATOMIC BONDING MODELS

We have seen that as far as solid state theory is concerned only the upper energy bands (valence bands) are of interest, since electrons at lower levels practically do not take part in interactions among atoms. This means that only electrons with greater than the Fermi energy (The energy corresponding to the highest filled electron filled state at 0 K in metal is called Fermi energy and designated by E_F) may be acted on and accelerated in the presence of an electric field. These are the electrons that participate in the conduction process, which are termed *free electrons*. As mentioned earlier that another charge entity called a *hole* is found in semiconductors and insulators. Holes have energies less than Fermi energy, E_F and also participate in electronic conduction. Thus the ensuing discussion reveals, the electrical conductivity is a direct function of the number of free electrons and holes. We may also note that the distinction between conductors, semiconductors and insulators lies in the number of these free electron and hole charge carriers.

Metals

For an electron to become free, it has to be excited or promoted into one of the vacant or empty and available energy states above Fermi energy, E_F (Fig. 14.3(a) and (b)). In Figs. 14.3(a) and 14.3(b), there are vacant energy states adjacent to the highest filled state at E_F . Obviously, very little energy is required to promote electrons into the low-lying empty states (Fig. 14.4). Generally, the energy supplied by an electric field is sufficient to excite large numbers of electrons into these conducting states.

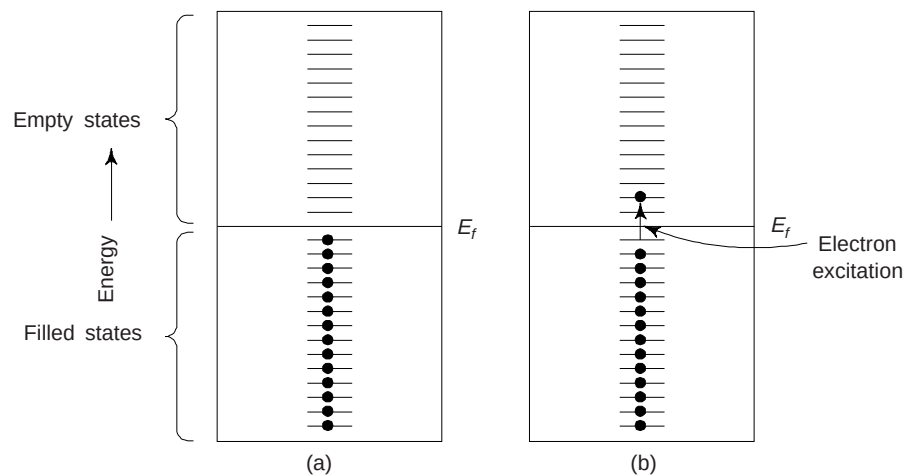


Fig. 14.4 Occupancy of electron states (a) before and (b) after an electron excitation for a metal

In a metallic bonding model it is assumed that all the valence electrons have freedom of motion and form an ‘*electron gas*’, which is uniformly distributed throughout the lattice of ion cores. Although these electrons are not locally bound to any particular atom, nevertheless, they must experience some excitation to become conducting electrons that are free in true sense. Obviously, only a fraction are excited, this still gives rise to a relatively large number of free, i.e. conducting electrons and, consequently, metals exhibit a high conductivity.

Semiconductors and Insulators

We have seen that for insulators and semiconductors, empty states adjacent to the top of the filled valence band are not available. To become free, we have to promote electrons across the energy band gap and into empty states at the bottom of the conduction band. This is possible only by supplying to an electron the difference in energy between these two states, which is approximately equal to the band gap energy, E_g (Fig. 14.5).

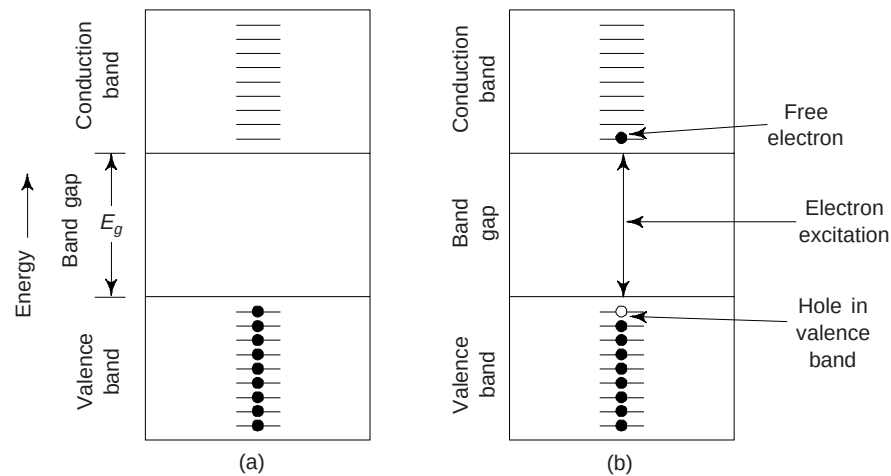


Fig. 14.5 Occupancy of electron states for an insulator and or semiconductor (a) before and (b) after an electron excitation from the valence band into the conduction band, in which both a free electron and hole are generated

The number of electrons excited thermally, i.e., by heat energy into the conduction band depends on the energy band gap width as well as temperature. At a given temperature, the larger the band gap energy E_g , the lower is the probability that a valence electron will be promoted into an energy state within the conduction band; this results in very few conduction electrons. This means that larger the band gap, the lower is the electrical conductivity at a given temperature. Obviously, the distinction between semiconductors and insulators lies in the width of the band gap; for semiconductors it is narrow, whereas for insulators it is relatively wide.

With the rise in temperature, the thermal energy that is available for electron excitation increases. Obviously, more electrons are promoted into the conduction band which gives rise to an enhanced conductivity.

Let us now examine the conductivity of insulators and semiconductors in the light of atomic bonding models. For electrically insulating materials, interatomic bonding is ionic or strongly covalent. Obviously, the valence electrons are tightly bound to or shared with the individual atoms. This means that these electrons are highly localized and are not in any sense free to wander throughout the crystal. In case of semiconductors, the bonding is covalent (or predominantly covalent) and relatively weak. This means that the valence electrons in semiconductors are not as strongly bound to the atoms. Consequently, these electrons are more easily freed by thermal excitation than they are for insulators.

7. ELECTRICAL RESISTIVITY OF METALS

Most metals are extremely good conductors of electricity. Room temperature conductivities of few common metals are given in Table 14.3. Metals have high conductivities due to the large number of free electrons that have been excited into empty states above the Fermi energy. Obviously, n has a large value in the conductivity expression (14).

Table 14.3 Room-temperature electrical conductivity for few metals and alloys

<i>Metal or alloy</i>	<i>Electrical conductivity ($\Omega\text{-m}$)⁻¹</i>
Copper	6.0×10^7
Silver	6.8×10^7
Gold	4.3×10^7
Aluminium	3.8×10^7
Iron	1.0×10^7
Platinum	0.94×10^7
Brass (70 Cu – 30 Zn)	1.6×10^7
Plain carbon steel	0.6×10^7
Stainless steel	0.2×10^7

Let us now discuss conduction in metals in terms of the resistivity (the reciprocal of conductivity).

The crystalline defects serve as scattering centers for conduction electrons in metals and increase in their number raises the resistivity, i.e. lowers the conductivity. The concentration of these imperfections depends on temperature, composition, and the degree of cold work of a metal specimen. It has been observed experimentally that the total resistivity of a metal is the sum of the contributions from the thermal vibrations, impurities and plastic deformation; i.e., the scattering mechanism act independently of one another. Mathematically, we can write this as follows:

$$\rho_{\text{total}} = \rho_t + \rho_i + \rho_d \quad (15)$$

where ρ_t , ρ_i and ρ_d are the individual thermal, impurity and deformation resistivity contributions, respectively. Equation (15) is sometimes called as *Matthiessen's rule*. Figure 14.6 shows a plot of resistivity versus temperature for copper and several copper-nickel alloys in annealed and deformed states. Figure also show the influence of each ρ variable on the total resistivity. The additive character of the individual resistivity contributions is demonstrated at -100°C .

7.1 Factors Affecting Resistivity

(i) *Influence of Temperature:* Any rise in temperature of a conductor (which contains small amounts of impurities) increases thermal agitation of the metallic ions as they vibrate about their mean position. This reduces the mean free path and restricts the free movement of electrons, thus reducing the conductivity of the metal, i.e. this increases the resistivity of metal. For the pure metal and all the copper-nickel alloys shown in Fig. 14.6, the resistivity rises linearly with temperature above about -200°C . Thus

$$\rho_t = \rho_o + aT \quad (16)$$

where ρ_o and a are constants for each particular metal.

(ii) *Influence of Impurities:* Another factor which reduces the mean free path of electrons is the impurity or solute atoms. The solute atoms provide the breakage in the regular crystalline structure, thus presenting an obstacle in the movement of electron waves. A solid solution alloy will always have lower conductivity than its pure components though both individual components have higher conductivity than the alloy. For additions of a single impurity that forms a solid solution, the impurity resistivity ρ_i is related to the impurity concentration C_i in terms of the atom fraction (at %/100) as follows:

$$\rho_i = AC_i (1 - C_i) \quad (17)$$

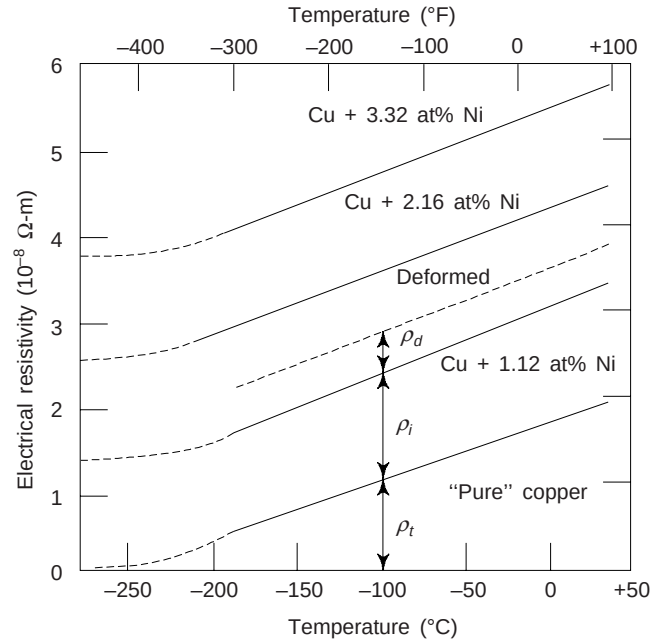


Fig. 14.6 The electrical resistivity vs. temperature curves for Cu and three copper-nickel alloys, one of which has been deformed. The contributions to the resistivity due to thermal, impurity and deformation are shown at -100°C

where A is a composition-independent constant that is a function of both the impurity and host metals. The influence of Ni impurity additions at about room temperature resistivity of Cu is shown in Fig. 14.7, upto 50 Wt% Ni; over this composition range Ni is completely soluble in Cu.

We may note that Ni atoms in Cu act as scattering centres, and increasing the concentration of Ni in Cu results in the enhancement of resistivity.

One can use the rule of mixtures expression for a two-phase alloy consisting of α and β phases to approximate the resistivity as follows:

$$\rho_i = \rho_{\alpha}V_{\alpha} + \rho_{\beta}V_{\beta} \quad (18)$$

where the V 's and ρ 's represent volume fractions and individual resistivities for the respective phases.

(iii) *Influence of Plastic Deformation:* Plastic deformation also raises the electrical resistivity as a result of increased number of electron-scattering dislocations. The effect of plastic deformation on resistivity is also shown in Fig. 14.6.

(iv) *Effect of Pressure:* At room temperature the general behaviour of ρ of metal is to decrease initially with increasing pressure and it may pass through a minimum. The initial decrease is due to the effect of pressure in reducing the amplitude of lattice vibrations. The subsequent increase is probably due to modification of the electron band structure which leads to increased phonon scattering.

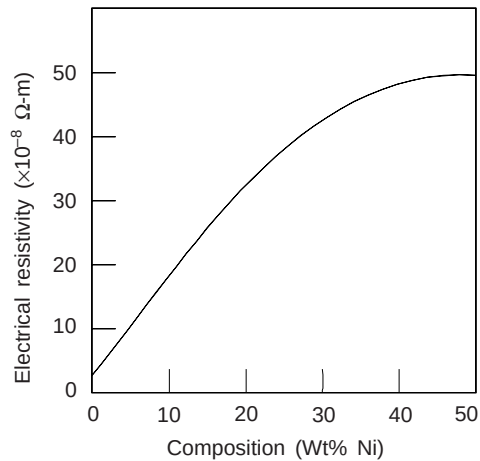


Fig. 14.7 Room temperature electrical resistivity vs. composition for Cu-Ni alloys

8. ELECTRICAL CHARACTERISTICS OF ALLOYS USED FOR COMMERCIAL PURPOSES

Copper is the most widely used metallic conductor due to its electrical and other properties. For many electrical applications, oxygen-free-high-conductivity (OFHC) copper having extremely low oxygen and other impurity contents is produced. Aluminium, having about one-half conductivity that of copper, is also frequently used as an electrical conductor. Silver has higher conductivity than either Cu or Al, however, its use is restricted due its very high cost compared to Cu and Al.

On occasion, it is necessary to improve the mechanical strength of a metal alloy without impairing significantly its conductivity. Both *solid solution alloying* and *cold working* improve strength at the expense of conductivity, and thus a trade off must be made for these two properties. Most often strength is enhanced by introducing a second phase that does not have so adverse an effect on conductivity, e.g. copper-beryllium alloys are precipitation hardened, but even so, the conductivity is reduced by about a factor of 0.75 over high purity Cu.

9. MECHANISMS OF STRENGTHENING IN METALS

We often need to design alloys having high strengths yet some ductility and toughness; ordinarily, ductility is sacrificed when an alloy is strengthened. Almost all strengthening techniques rely on this simple principle: *restricting or hindering dislocation motion renders a material harder and stronger*. The strengthening mechanisms for single phase metals are:

(i) *Strengthening by Grain-size Reduction*: A fine grained material (one that has small grains) is harder and stronger than one that has coarse grained, since the former has a greater total boundary area to impede dislocation motion. For many materials, the yield strength σ_y varies with grain size in accordance with

$$\sigma_y = \sigma_o + k_y d^{-1/2} \quad (19)$$

where d is the average grain diameter, and σ_o and k_y are constants for a particular material. Equation (19) is termed the *Hall Petch equation*. We may note that Eq. (19) is not valid for both very large (i.e., coarse) grain and extremely fine grain polycrystalline materials.

The grain size reduction improves not only strength but also the toughness of many alloys.

(ii) *Solid-Solution Strengthening*: Another technique to strengthen and harden metals is alloying with impurities that go into either substitutional or interstitial solid solution. This is called solid-solution strengthening. Alloying elements or impurities increase the resistivity of materials due to the presence of solute atoms in solid solutions. The solute atoms interfere with the motion of electrons. In a Cu-Ni alloy, the Cu atoms do not impede the motion of free electrons but Ni atoms impede their motion. We may note that high-purity metals are always softer and weaker than alloys composed of the same base metal. Increasing the concentration of the impurity results in an attendant increase in tensile and yield strength. Alloys are stronger than pure metals because impurity atoms that go into solid solution normally impose lattice strains on the surrounding host atoms.

(iii) *Strain Hardening or Work Hardening or Cold Working*: This is the phenomenon whereby a ductile metal becomes harder and stronger as it is plastically deformed. Sometimes it is also called *work hardening*, or, because the temperature at which deformation takes place is “cold” relative to the absolute melting temperature of the metal, *cold working*. Most metals strain harden at room temperature.

Cold work increases the resistivity of the metal as it changes the crystal structure and the localised strain interfaces with the movement of electron. However, subsequent annealing decreases the resistivity. Hand drawn copper wire has therefore high resistivity than annealed copper.

It is sometimes convenient to express the degree of plastic deformation as *percent cold work* than as strain. Percent cold work is

$$\% CW = \left(\frac{A_o - A_d}{A} \right) \times 100 \quad (20)$$

where $A_o \rightarrow$ original area of cross-section that experiences deformation, and $A_d \rightarrow$ the area after deformation.

Strain hardening is often utilized commercially to enhance the mechanical properties of metals during fabrication procedures.

We may note that deformation and strengthening of multiphase alloys are more complicated, involving concepts which are beyond the scope of this book.

Example 1 Calculate the resistance of an aluminium wire 100 m long and having cross-sectional area of 3 sq. mm at 20°C. Given, the resistivity of Al at 20°C = $2.66 \times 10^{-8} \Omega\text{-m}$. [AMIE]

Solution Given

$$\begin{aligned} l &= 100 \text{ m} \\ \rho &= 2.66 \times 10^{-8} \Omega\text{-m} \\ A &= 3 \times 10^{-6} \text{ sq. m} \end{aligned}$$

$$\begin{aligned} R &= \frac{\rho l}{A} \\ &= \frac{2.66 \times 10^{-8} \times 100}{3 \times 10^{-6}} \\ &= 88.67 \times 10^{-2} \Omega. \end{aligned}$$

Example 2 The resistivity of pure copper is 1.56 micro-ohm-cm. An alloy of copper containing 2 atomic percent nickel has a resistivity of 4.06 micro-ohm-cm. An alloy of copper, containing 1 atomic percent silver, has a resistivity of 1.7 micro-ohm-cm. Find the resistivity of a copper alloy containing 1 atomic percent nickel and 3 atomic percent silver. [Diploma]

Solution Resistivity of pure Cu = 1.56 micro-ohm-cm. Resistivity of Cu containing two atomic percent Ni = 4.06 micro-ohm-cm

$$\begin{aligned} \therefore \text{Increase due to one atomic \% Ni} &= \frac{4.06 - 1.56}{2} \\ &= 1.25 \text{ micro-ohm-cm} \end{aligned}$$

Now, resistivity of copper, containing one atomic percent silver = 1.7 micro-ohm-cm

$$\begin{aligned} \therefore \text{Increase in resistivity due to one atomic \% Ag} &= 1.7 - 1.56 \text{ micro-ohm-cm} \\ &= 0.14 \text{ micro-ohm-cm} \end{aligned}$$

Resistivity of copper alloy containing one atomic percent Ni and 3 atomic percent

$$\begin{aligned} \text{Ag} &= 1.56 + 1.25 + 3 \times 0.14 \\ &= 1.56 + 1.25 + 0.42 \\ &= 3.23 \text{ micro-ohm-cm} \end{aligned}$$

Example 3 The resistivity of pure copper at room temperature is $1.8 \times 10^{-8} \Omega\text{-m}$. The resistivity of Cu – 4% Ni alloy at room temperature is $7 \times 10^{-8} \Omega\text{-m}$. Show that the resistivity due to impurity scattering per % of Ni in the Cu lattice is $1.3 \times 10^{-8} \Omega\text{-m}$. [Diploma]

Solution Let us take that the resistivity of copper at 0 K is negligible. Now, $1.8 \times 10^{-8} \Omega\text{-m}$ is the thermal part of resistivity of copper as well as of the alloy between 0 K and room temperature.

$$\therefore \text{Impurity scattering per \% of Ni is then} = \frac{(7.0 \times 1.8) \times 10^{-8}}{4} = 1.3 \times 10^{-8} \Omega\text{-m}$$

10. INSULATORS

We have read that in an ideal insulator, all valence electrons are occupied in bond formation and almost no free electrons are available for electrical conduction. Thus an insulator (electrical) is a non-metallic

material that has a filled valence band at 0 K and relatively wide energy band gap. Consequently, the room temperature electrical conductivity is very low, less than about $10^{-10} (\Omega\text{-m})^{-1}$. A good insulator may have a resistivity as high as $10^{14} (\Omega\text{-m})$. Common electrical insulating materials are polyethylene, bakelite, lucite, mica, PVC, rubber, porcelain etc. Most polymers and ionic ceramics are insulating materials at room temperatures. Thermal agitation and imperfections adversely affect the insulating properties of a material but, as is real materials, a few free electrons are always there to conduct electricity.

Non-metallic crystals are held together by ionic and covalent bonds and valence electrons are much more closely associated with their atoms than in the metallic bond. Generally, insulators are those materials which have the electrons completely filled in the *Brillouin Zone* and also at the same time have wide energy gaps.

The conductivity of insulators increases with the rise in temperature because more and more conducting electrons are released. However, the materials which are insulators at 0 K but develop significant conductivities at room temperature become semiconductors. Room temperature conductivities of few selected non-metallic materials are given in Table 14.4. Of course, many insulating materials are used on the basis of their ability to insulate, and thus a high electrical resistivity is desirable.

Table 14.4 Room temperature conductivities for some selected non-metallic materials

Material	Electrical conductivity [[$\Omega - m$] ⁻¹]	Material	Electrical conductivity [[$\Omega - m$] ⁻¹]
Graphite	$3 \times 10^4 - 2 \times 10^5$	POLYMERS	
CERAMICS		Phenol-formal-dehyde	10^{-9} - 10^{-10}
Concrete (Dry)	10^{-9}	Polymethyl-methacrylate	$< 10^{-12}$
Soda lime glass	10^{-10} - 10^{-11}	Nylon 6, 6	10^{-12} - 10^{-13}
Porcelain	10^{-10} - 10^{-12}	Polystyrene	$< 10^{-14}$
Borosilicate glass	$\sim 10^{-13}$	Polyethylene	10^{-15} - 10^{-17}
Aluminium oxide	$< 10^{-13}$	Polytetra fluoroethylene	$< 10^{-17}$
Fused silica	$< 10^{-13}$		

One can classify the requirements of good insulating materials as electrical, mechanical, thermal and chemical. Electrically the insulating material should have high resistivity to reduce the leakage current and high dielectric strength so that it may withstand higher voltage without being punctured or broken down. Moreover, the insulator should have small dielectric loss.

Insulators are used on the basis of volume and not weight and hence a low density is preferred. A uniform viscosity and for liquid insulators ensures uniform thermal and electrical properties.

Liquid and gaseous insulators are used also as coolants, e.g. transformer oil, hydrogen and helium are used for insulation and cooling purposes. Obviously, for such materials good thermal conductivity is a desirable property. To prevent mechanical damage, the insulator should also have small thermal expansion. Moreover, it should be non-ignitable or if ignitable it should be self extinguishable.

Chemically, the insulating materials should be resistant to oils, liquid gas fumes, acids and alkalies. Moreover, the insulator should not absorb water particles as water lowers the insulation resistance and the dielectric strength.

Insulators should have certain mechanical properties depending on the use to which they are put. Thus when insulators are used for electric machine insulation, they should have sufficient mechanical strength to withstand vibration. In such cases, good heat conducting property is also desirable.

In making dielectric capacitors insulating materials with large electronic and ionic polarizabilities and therefore large permittivity are used. Titanium oxide with a permittivity ~ 100 is a good example of such a material. We may note that the use of molecules with a permanent dipole moment is not desirable because of possibility of large dielectric losses at high frequencies.

Properties of the some of the insulating materials are given in Table 14.5.

Table 14.5 Properties of few insulating materials at room temperature

<i>Material</i>	<i>Dielectric constant (ϵ_r)</i>	<i>Dielectric strength (kV/mm)</i>	<i>Density ($\times 10^{-3}$) (kg/m³)</i>	<i>Resistivity (ρ) ($\Omega\cdot m$)</i>
Mica	4-5.5	60-125	2.5-2.7	10^{14} - 10^{19}
Glass	4-10	20-30	2.2-4.0	10^{13} - 10^{16}
Ebonite (RP)	4-4.5	25	1.3	1×10^{20}
Bakelite	4-4.6	10-40	1.2	—
Asbestos	—	2	2.3-2.6	2×10^7
Rubber (soft)	2.6-3.0	15-25	1.7-2.9	4×10^{15}
Silk (natural)	4.5	—	—	—
Polystyrene	2.2-2.8	25-50	1.05-1.65	5×10^{13} – 5×10^{19}
Polyvinyl chloride (PVC) resin	3.1-3.5	50	1.38	—

11. DIELECTRICS

These are the materials or insulators which have the unique characteristic of being able to store electric charge. The electrons in these substances are localized in the process of bonding the atoms together. Obviously, covalent or ionic bonds, a mixture of both, or Vander Waals bonding between closed shell atoms give rise to solids or gases which exhibit dielectric or insulating properties. Dielectric materials may be gases, liquids or solids with the exception of air which is the insulating material between the bare conductors of the overhead electric grid system. Liquid dielectrics are used mainly as impregnants for high voltage paper insulated cables and capacitors as filling and cooling media for transformers and circuit breakers. Most common properties of dielectric materials are: (i) dielectric constant (ii) dielectric strength (iii) insulation resistance (iv) surface resistivity (v) loss factor (vi) tangent of loss factor in terms of a capacitor or phase difference (vii) polar and non-polar materials.

Materials, which are capable of separating electrical conductors circuit breakers, e.g. silicon, oils, liquid dielectrics have high dielectric constant, high resistance, high dielectric strength when moisture and impurities are removed from them. They have high temperature dissipation capacity and least dielectric losses. Their dielectric constant is greater than one. Following materials are important from engineering point of view:

Mica is the widely used insulating material in switch gears armature windings, electrical heating devices like iron, hot plates etc. It is also used in capacitors for high frequency application. Mica is an inorganic compound of silicates of aluminium, soda potash and magnesia. It is crystalline in nature and can be easily split into very thin flat sheets. The two important types of mica are: (i) muscovite and (ii) phlogopite. Mica has a good dielectric strength and mechanical strength. Its dielectric constant varies between 5 and 7.5, loss tangent between 0.0003 and 0.015 and dielectric strength between 700 and 1000 kV/mm.

Asbestos is also used as an insulator in the form of paper, tape, cloth and board. Asbestos is widely used in panel boards, insulating tubes and cylinders in the construction of air cooled transformers. Asbestos is an inorganic material, which is used to designate a group of naturally occurring fibre material. Asbestos has good dielectric and mechanical properties.

Ceramics are generally non-metallic inorganic compounds, e.g. silicates, aluminates, oxides, carbides, borides, nitrides and hydroxides. Ceramics used as dielectrics may be broadly described as alumina, porcelains, ceramics, titanates, etc. These have excellent dielectric and mechanical properties. The dielectric constant of most commonly used ceramics varies between 4 and 10. These are used in switches in plug holders, thermocouples, cathode heaters, vacuum type ceramic metal seals etc. Ceramic capacitors may be operated at high temperatures and can be moulded into any shape and size.

Electric grade ceramics are used for the manufacturing of insulators, terminal blocks, plates, frames, coils, etc. They must have low losses, good insulating properties and high strength.

Insulators for operation at low frequencies are made of electric-grade porcelain which possesses fairly good electric properties. A drawback is that it has high losses which increase sharply on heating above 200°C, and a low strength.

Insulating parts for operation at high frequencies are mostly made of steatite, a talc base material. Steatites contain no harmful impurities and retain their properties at temperatures upto 100°C. They can be pressed easily, shrink on burning by only 1-2%, and are suitable for making parts in which dense but porous structure and accurate dimensions are essential. In contrast to other kinds of ceramics, steatite can be cut quite easily (after burning). Among their drawbacks are cracking under the action of rapidly varying temperatures and some difficulties the burning involves.

Glass is an inorganic insulating material, which comprises of complex system of oxides. Silica (SiO_2) is the most essential constituent of many commercially used glasses. It is fused with alkali (like potash, soda etc.) and some base (like lime, lead oxide etc.). The silica glass (having 100% SiO_2) is the best insulating material. The dielectric constant of glass varies between 3.7 and 10 and loss tangent between 0.0003 and 0.01 and dielectric strength between 2.5 and 50 kV/mm. Glass is used in electric bulbs, X-ray tubes, mercury switches, electronic valves as insulating material. It is also used in capacitors as dielectric material.

Resins are organic polymers and may be natural or synthetic. The synthetic resins are produced artificially. The commonly used synthetic resins are polyethylene, polystyrene, polyvinyl chloride, acrylic resins, teflon, nylon, etc. These have good dielectric and mechanical properties. The dielectric constant of resins varies between 2 and 4.5, the loss tangent between 0.0002 to 0.04 and dielectric strength is quite high. These are used in transformers, high frequency capacitors. These are also used as a dielectric material in d.c. capacitors.

Rubber

These are organic polymers and may be natural or synthetic. The natural rubber obtained from rubber tree has limited applications due to its resistance to low and high temperatures. The synthetic rubbers are produced artificially by *copolymerisation* of isobutylene and isoprene. These have good electrical and thermal properties. The dielectric constant of rubber varies between 2.5 and 5, and loss tangent between 0.01 and 0.03. Rubber's are used as an insulating material for electric wires, tapes, cables, coatings, motor windings, transformers, etc.

Gaseous dielectrics, e.g. nitrogen, hydrogen, etc. have dielectric constant 1.0.

The best dielectrics are polystyrene, polycarbonate, polyethylene, polymide, mineral oil, pure alumina and pure silica.

11.1 Dielectric Constant

In an electric field, the charge density D is directly proportional to the applied field, i.e.

$$D = \epsilon E \quad (21)$$

where ϵ is the dielectric constant or permittivity of the material placed between the electrodes. For vacuum,

$$D = \epsilon_0 E \quad (21(a))$$

where $\epsilon_0 = 8.854 \times 10^{-12}$ farad/m is called the absolute permittivity.

One can define the relative permittivity (ϵ_r) of a dielectric as:

- (i) The ratio of electric field density produced in the dielectric medium (E_0) to that produced in vacuum (E) by the same electric field strength, i.e.

$$\epsilon_r = \frac{E_0}{E}$$

ϵ_r is also called as macroscopic dielectric constant.

- (ii) The ratio of capacitance of a condenser containing a given dielectric to the same condenser with vacuum as dielectric.

Thus relative permittivity can be express as

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \quad (22)$$

Now, charge density = $\epsilon_0 \epsilon_r E = \epsilon_0 E + \epsilon_0 (\epsilon_r - 1) E$

where $\epsilon_r - 1 = \chi$ is called electric susceptibility. $\chi = \text{Bound charge density/free charge density}$.

Permittivity is influenced by permeability (μ), stress distribution, temperature and frequency.

11.2 Phase Difference and Dielectric Loss

A dipole tends to align itself along the direction of applied electric field and for ac fields tends to follow the field and be in a phase with it. However, the interaction of this dipole with other dipoles in the medium prevents this and this leads to dielectric loss. This loss appears as heat. This dielectric loss is connected with ϵ_r'' , the imaginary part of the dielectric constant. The rate of loss of energy in unit volume of the material is obtained as

$$W = \frac{1}{2} \epsilon_0 \epsilon_r'' \omega E_0^2 \quad (23)$$

Obviously, the energy loss is proportional to ϵ_r'' . Usually the dielectric loss is expressed in terms of a quantity called 'the loss tangent', $\tan \delta$. This is defined as $\tan \delta = \epsilon_r''/\epsilon_r'$. The angle δ is the compliment of angle ϕ , i.e. $90 - \phi = \delta$, where ϕ is the angle between the applied field and the resultant current vector (Fig. 14.8).

From alternating current studies, we have

$$\text{Power} = V_R I_R \cos \phi$$

where V_R and I_R are rms values of voltage and current, and for a circuit containing capacitance C and resistance R . We have

$$\tan \delta = \frac{1}{\omega CR} \quad (24)$$

Using (22), one obtains for a dielectric

$$W = \frac{1}{2} \epsilon_0 \epsilon' \tan \delta \omega E_0^2 = \frac{1}{2} \frac{E_0^2}{\rho} \quad (25)$$

where ρ is the resistivity. Thus

$$\tan \delta = \frac{1}{\omega \epsilon_0 \epsilon' \rho} \quad (26)$$

The energy losses in a dielectric material are due to: (i) ionization, (ii) leakage current, (iii) polarization, and (iv) structural inhomogeneity.

Ionization losses occurs in gases and solids having pores with entrapped gases. With the rise in field strength applied to a gas, a stage is reached when the gas molecules gets ionized due collisions. This leads to enhanced conduction leading to dielectric losses. At low or ordinary electric fields, the conductivity is low and hence the loss is also small (Eq. (25)). For example, for $\rho \approx 10^{16} \Omega\text{-m}$, $\epsilon_r' \approx 1$ and at 100 Hz, $\tan \delta < 2 \times 10^{-8}$. In case of solids with gas inclusions the loss tangent increases with voltage from the voltage required for the ionization of gas molecules to the value of voltage at which ionization is complete. When such inclusions are high, there are chances of the failure of the dielectric.

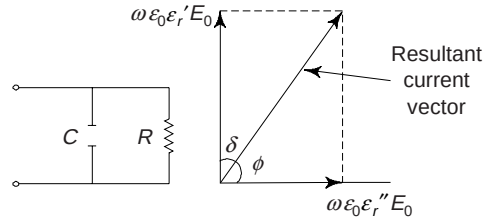


Fig. 14.8 Vector relationship between the field vector E_0 , the current vector $\omega \epsilon_0 \epsilon_r' E_0$ which leads the field by 90° and the current vector $\omega \epsilon_0 \epsilon_r'' E_0$ which is in phase with the field. The angle δ is shown

Due to the leakage current, the losses in highly conducting liquids and solids are high (Eq. (24)). In case of neutral liquids the losses will be small as their conductivity is low. Purity of oil is quite important as otherwise dielectric losses increase due to conduction. Pure transformer oil has $\tan \delta \approx 0.0005$.

Dipole losses in the radiofrequency region are usually due to dipole rotation or ions jumping from one equilibrium position to another. The dielectric losses associated with ions, the frequencies of which fall in the infrared region, are usually referred to as *infrared absorption*. Similarly, the losses in the optical region, associated with the electrons, are referred to as *optical absorption*.

11.3 Dielectric Strength

As the voltage applied across a dielectric is increased, dielectric loses its insulation property at a certain voltage called the *breakdown voltage* (V_{br}). The corresponding field strength (E_{br}) is given by

$$E_{br} = \frac{V_{br}}{t} \quad (27)$$

Here t is the thickness of the material. This voltage, V_{br} per unit thickness of the material is called the *dielectric strength* and usually expressed in Volts/mil or kV/millimeter or mega Volt/m. The breakdown of a dielectric is due to the collision of accelerated electrons or ions with molecules. At relatively high fields, the electrons in the dielectric gain enough energy to knock other charged particles and make them available for conduction.

Dielectric failure involves deterioration and electric breakdown or cascading. Resistance to electronic breakdown in a dielectric is *intrinsic dielectric strength* of the material. Intrinsic breakdown in a dielectric is caused by imperfection. When the applied potential across the dielectric becomes high enough, a few electrons are broken loose at points where their bonds are strained by the presence of imperfections. Upon being freed, these electrons are accelerated rapidly through the material. The dielectric material loses completely its insulating capacity as a result of this process.

Obviously, dielectric strength is structure sensitive property of the material. We may note that the actual thickness of the material affects the breakdown potential (V_{br}) per unit thickness, i.e. dielectric strength. Usually thicker materials possess lower dielectric strength than thinner materials.

The fundamental breakdown mechanisms in a solid are: (i) *intrinsic breakdown* (ii) *thermal breakdown* (iii) *discharge breakdown* and (iv) *electrochemical breakdown*.

11.4 Power Factor

The ratio of power loss in a material to the product of applied voltage and current is called power factor. The power factor and loss angle depend upon the nature of the material, applied voltage, humidity, temperature and frequency.

11.5 Polar and Non-Polar Materials

Permanent dipole moments exist in some molecules by virtue of an asymmetrical arrangement of positively and negatively charged regions. Such molecules are termed polar molecules. For example, in the HCl molecule (Fig. 14.9a), the electron of hydrogen atom spends more time moving around the Cl atom than around the H atom. Therefore the centre of negative charges does not coincide with that of positive charges, and the molecule has a dipole moment directed from the Cl atom to the H atom. That is, we may write H^+Cl^- . The electric dipole of HCl molecule is $p = 3.43 \times 10^{-30}$ C-m. In the CO molecule Fig. 14.9 (b), the charge distribution is only slightly asymmetric and the electric dipole moment is relatively small, about 0.4×10^{-30} C-m, with the carbon atom corresponding to the positive and the oxygen atom to the negative end of the molecule.

Polar molecules can also induce dipoles in adjacent non polar molecules, and a bond will form as a result of attractive forces between the two molecules. Moreover, the magnitude of this bond will be greater than for fluctuating induced dipoles.

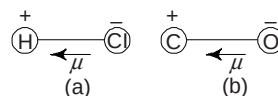


Fig. 14.9 Polar diatomic molecules

Vander Waals forces will also exist between adjacent polar molecules. We may note that the associated bonding energies are significantly greater than for bonds involving induced dipoles.

The strongest secondary bonding type, the hydrogen bond, is a special case of polar molecule bonding. We find that it occurs between molecules in which hydrogen is covalently bonded to fluorine (as in HF molecule), oxygen (as in H_2O molecule), and nitrogen (as in NH_3 molecule). For each H-F, H-O or H-N bond, the single hydrogen electron is shared with the other atom. Obviously, the hydrogen end of the bond is essentially a positively charged bare proton that is unscreened by any electrons. This highly positively charged end of the molecule is capable of a strong attractive force with the negative end of an adjacent molecule (Fig. 14.10). Clearly, this single proton forms a bridge between two negatively charged atoms. We may note that the magnitude of the hydrogen bond is generally greater than that of the other types of secondary bonds and may be as high as 0.52 eV/molecule ($= 51 \text{ kJ/mol}$). We find that the melting and boiling temperatures for hydrogen fluoride and H_2O are abnormally high in light of their low molecular weights, as a consequence of hydrogen bonding.

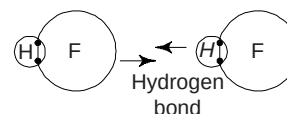


Fig. 14.10 Representation of hydrogen bonding in HF molecule

Polarization

When a material is placed in an electric field (Fig. 14.11), e.g. between the plates of a condenser, the field strength of charged particles within the material interact with the electric field. If the material is a conductor, some of the free electrons simply move to the side nearest the positive electrode until they totally counteract the applied electric field. Obviously, no field is left within the material. The displacement of charged particles occurs almost simultaneously, bringing about equilibrium.

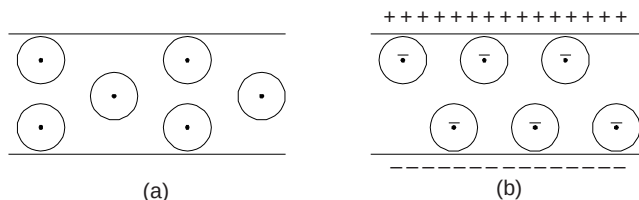


Fig. 14.11 Electronic polarization (a) In the absence of field (b) In the presence of electric field, the formation of induced dipole

If the material is non-conducting or insulator (dielectric), electrons can only be displaced locally as they are bound to the individual atoms. This local displacement of electrons, however, is sufficient to polarize the material. The negative electron cloud is displaced in each atom relative to the positive nucleus, thereby creating a small induced dipole whose negative pole is toward the positive side of the electric field. All dielectric materials are subjected to such *electronic polarization* (Fig. 14.11).

Induced polarization also occurs in ionic materials. In an ionic crystal, for e.g. the negative ions are attracted towards the positive side and vice versa as illustrated in Fig. 14.12. NaCl is a good example of ionic polarization. Its interionic separation is 0.1 nm and dipole moment is of the order of 10^{-29} C-m .

There are many molecular structures which have permanent electronic dipoles due to asymmetric nature of their atomic structures. Few examples of such molecules are: H_2O , HCl , polyvinylchloride, etc. Some molecules also contain permanent charges which produce dipole moments in the individual molecules. There are many polymers of this type (polymers) and glasses contain dipoles. The permanent molecular dipoles in such materials can rotate about their axis of symmetry to align with an applied field which exert a torque in them. This is known as *orientation polarization*. This type of polarization is rarely perfect as thermal fluctuations do not permit all the dipoles to align with the applied field. With the rise in temperature

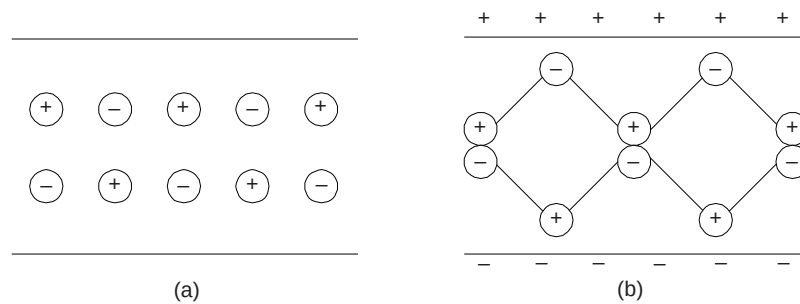


Fig. 14.12 Ionic Polarization (a) In the absence of external field (b) Ions displacement towards opposite charged plates in an external field

the degree of polarization decreases. In the process of orientation polarization carriers collect at the boundaries between the constituent materials resulting in a local distortion of the applied field. This is called as *Space-Charge polarization*.

The dipole moment of a single pair of charges is given by

$$\mu = qd \quad (28)$$

where q is one of the charges and d is the distance between charges.

Non-Polar Materials

Most of the hydrocarbons are non-polar. In non-polar materials the molecules, which are usually diatomic and composed of two atoms of the same type, may be represented as a positive nuclei of charge q surrounded by a symmetrically distributed negative electron cloud of charge $-q$. In the absence of an applied electric field the centres of gravity of the positive and negative charge distributions coincide. When the molecules are subjected to an external electric field the positive and negative charges experience electric forces tending to move them apart in the direction of external electric field. The distance moved is very small ($\sim 10^{-10}$ m) since the displacement is limited by restoring forces which increases with increasing displacement. The centres of positive and negative charges no longer coincide and the molecules are said to be polarized.

11.6 Insulation Resistance

This separates a number of conductors at different electrical potentials and does not allow a large flow of electric current between them. The difference of potential may cause leakage of current along two paths, i.e. (i) over the surface of insulation, and (ii) through the solid material of the insulators.

We may note that the resistance offered along the said two paths is not the same due to the nature of the material. The former is due to the surface resistivity of the material, whereas the latter is the result of the volume resistivity of the material. The combined effect of two is called the insulation resistance of the material.

We may define the surface resistivity as the resistance between the two opposite edges of a square of unit area of insulation surface. The volume resistivity is defined as the resistance presented to the flow of an electric current by a material of unit cross section and of unit length at 0°C .

The resistive properties of real dielectrics are:

- (i) Under the action of a potential, mobility with which any conducting species moves.
- (ii) Temperature variation of conductivity to state why the conductivity of a dielectric varies in exponential manner with temperature.
- (iii) To analyse the time response of dielectrics, frequency or time variation of conductivity.
- (iv) While designing an equipment, the knowledge of breakdown or the highest voltage a dielectric will withstand is of interest. Usually insulators breakdown at 10^6 V/m to 10^9 V/m dc at 20°C .

11.7 Classification of Dielectrics

One can classify dielectric as under following three groups:

<i>Class</i>	<i>Effect of electric field</i>
Simple dielectrics	Creates dipoles
Paraelectrics	Orients dipoles
Ferroelectrics	Orients domains of aligned permanent dipoles

11.8 Ferroelectricity

The group of dielectric materials called *ferroelectrics* exhibit spontaneous polarization i.e., polarization in the absence of an electric field. In a sense, ferroelectrics are the electric analog of the ferromagnets, which may display permanent magnetic behaviour. In ferroelectrics, the polarization can be changed and even reversed by an external electric field. The reversibility of the spontaneous polarization is due to the fact that the structure of a ferroelectric crystal can be derived from a non-polarized structure by small displacement of ions. In most ferroelectric crystals, this non polarized structure becomes stable if the crystal is heated above a critical temperature, the *ferroelectric Curie temperature* (T_c); i.e. the crystal undergoes a phase transition from the polarized phase (ferroelectric phase) into an unpolarized phase (Paraelectric phase). The change of the spontaneous polarization at T_c can be continuous or discontinuous. The T_c of different types of ferroelectric crystals range from a few degrees absolute to a few hundred degrees absolute.

From a practical standpoint ferroelectrics can be divided into two classes: (i) In this *first class* of ferroelectrics, polarization can occur only one crystal axis, e.g. Rochelle salt, KH_2PO_4 , $(\text{NH}_4)_2\text{SO}_4$, guanidine aluminium sulphate hexahydrate, glycine sulphate, colemanite and thiourea, (ii) In this *second class* of ferroelectrics, spontaneous polarization can occur along several axes that are equivalent in the paraelectric phase, e.g. BaTiO_3 -type (or perovskite type) ferroelectrics, $\text{Cd}_2\text{Nb}_2\text{O}_7$, PbN_2O_6 , certain alums, such as methyl ammonium alum and $(\text{NH}_4)_2 - \text{Cd}_3(\text{SO}_4)_3$.

One can also place ferroelectrics as *proper ferroelectrics* and *improper ferroelectrics*. In proper ferroelectrics, for e.g., BaTiO_3 , KH_2PO_4 , and Rochelle salt, the spontaneous polarization is the order parameter. In improper ferroelectrics, the spontaneous polarization can be considered a by-product of another structural phase transition. The properties of some representative ferroelectrics are given in Table 14.6.

Table 14.6 Properties of some representative ferroelectric crystals

<i>Group</i>	<i>Crystal</i>	<i>Curie temperature</i> T_c ($^{\circ}\text{K}$)	P_s $\text{C/m}^{-2} \times 10^{-2}$	<i>At T</i> ($^{\circ}\text{K}$)
Ilmenites and Perovskites	GeTe	670	—	—
	LiNbO_3	1480	71	296
	KNbO_3	710	30	600
	BaTiO_3	393	26	300
KDP type	SrTiO_3	32	3	4.2
	KH_2PO_4 (KDP)	123	4.7	100
	KD_2PO_4	213	5.5	100
	$\text{Rb}_2\text{H}_2\text{PO}_4$	147	5.6	90
	KH_2AsO_4	97	5.0	78
TGS type	$(\text{NH}_2\text{CH}_2\text{COOH})_3 \cdot \text{H}_2\text{SO}_4$ (Triglycine sulphate)	322	2.8	275
Rochelle salt type	$\text{NaKC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$	296 (upper)	0.25	275
	(Rochelle salt)	255 (lower)		

The spontaneous polarization in ferroelectrics can occur in at least two equivalent crystal direction, thus a ferroelectric crystal consists in general of regions of homogeneous *domains*. Ferroelectrics of the first

class consist of domains with parallel and antiparallel polarization (Fig. 14.13(a)), whereas ferroelectrics of the second class can assume much more complicated domain configurations (Fig. 14.13(b)). The region between two adjacent domains is called a *domain wall*. Within this wall, the spontaneous polarization changes its direction.

When an electric field is applied to a specimen of a ferroelectric crystal, the polarization first rises rapidly with applied field to a value above which the dependence is linear. Linear extrapolation to zero field gives, the *saturation* or *spontaneous polarization*. On subsequently reducing the field to zero, remanent (residual) polarization remains. The negative field to reduce the polarization to zero is called the *coercive field*.

Obviously, a ferroelectric crystal can be switched by the application of an electric field and a *hysteresis loop* is associated with the switching. The existence of a dielectric hysteresis loop in a dielectric material implies that the substance possesses a spontaneous polarization, P_s and the value of spontaneous polarization (depending upon the shape of hysteresis loop) depends upon a number of factors such as the dimensions of the specimen, the temperature, the texture of the crystal, the thermal and electrical properties of the crystal. A typical hysteresis loop is shown in Fig. 14.14. For most ferroelectrics, the values of P_s are between 10^{-7} and 10^{-4} C/cm². In non ferroelectric dielectrics, electric fields between 10^5 and 10^8 V/cm would be necessary in order to achieve such large polarizations.

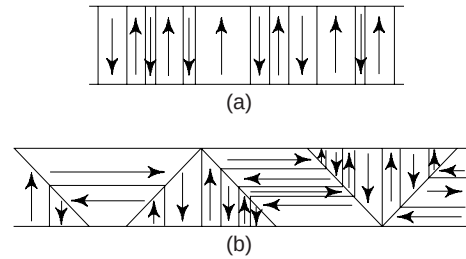


Fig. 14.13 Domain configurations (simplified) encountered in ferroelectric. (a) First class (b) Second class

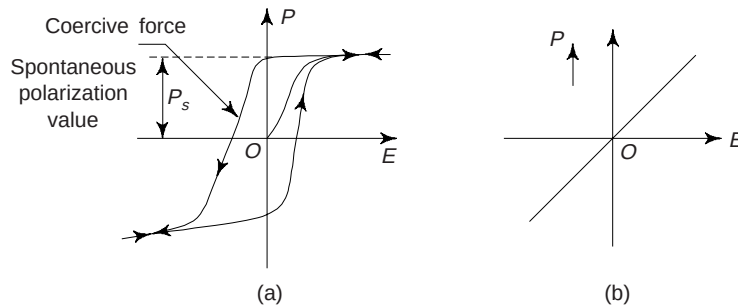


Fig. 14.14 Net polarization P of a ferroelectric crystal versus externally applied electric field. (a) Hysteresis (b) P - E relation above Curie temperature

The hysteresis loop of a ferroelectric material changes its shape as the temperature is increased. The height and width decrease with increase of temperature. At a certain temperature known as *ferroelectric Curie temperature* (T_c), the loop merges to a straight line and the ferroelectric behaviour of the material disappears. The electric susceptibility χ_E in the *paraelectric phase* (above Curie temperature, T_c) is related to the temperature by the Curie-Weiss law

$$\chi_E = \frac{C}{T - T_c} \quad (29)$$

where C is Curie constant and $T_c \rightarrow$ Curie-Weiss temperature.

As a rule, the dielectric constant ϵ measured along a ferroelectric axis increases in the paraelectric phase when the Curie Temperature, T_c is approached. In many ferroelectrics, this increase can be approximated by Curie-Weiss law,

$$\epsilon = \frac{C}{T - T_0} \quad 29(a)$$

where T_0 is equal or somewhat smaller than T_c . For BaTiO_3 , this law holds unaltered upto frequencies of 2.4×10^{10} Hz. Ferroelectric crystals will have an extremely large dielectric constant.

One of the most common ferroelectrics is the barium titanate (BaTiO_3). The spontaneous polarization in this crystal is a consequence of the positioning of the Ba^{2+} , Ti^{4+} , and O^{2-} ions within the unit cell (Fig. 14.15). The Ba^{2+} ions are located at the corners of the unit cell, which is of tetragonal symmetry (a cube that has been elongated slightly in one direction). The dipole moment results from the relative displacements of the O^{2-} and Ti^{4+} ions from their symmetrical positions as shown in the side view of the unit cell. We can see that O^{2-} ions are located near, but slightly below, the centers of each of six faces, whereas the Ti^{4+} ion is displaced upward from the unit cell corner. Obviously, a permanent ionic dipole moment is associated with each unit cell. However, when BaTiO_3 is heated above its *ferroelectric* T_c ($= 120^\circ\text{C}$), the unit cell becomes cubic, and all ions assume symmetric positions within the cubic unit cell; the material now has a perovskite crystal structure, and ferroelectric behaviour ceases. Ceramic materials and especially ferroelectrics are the most important ones among dielectrics. Ceramics may have very diversified electric properties (Table 14.7) and are almost insusceptible to ageing and heating.

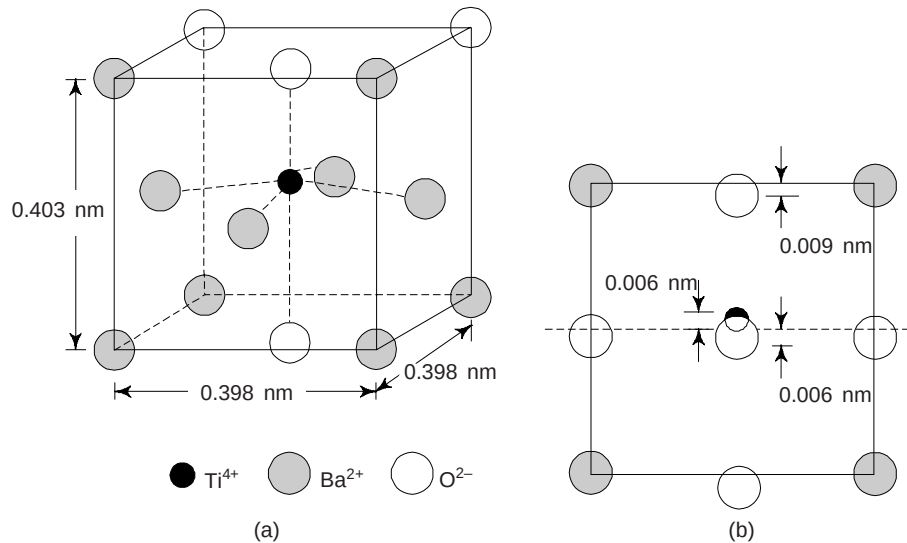


Fig. 14.15 BaTiO_3 unit cell (a) in an isomeric projection, and (b) looking at one face, exhibiting the displacement of Ti^{4+} and O^{2-} ions from the center of the face

Table 14.7 Properties of dielectrics

Material	ϵ	$\tan \delta \times 10^4$	ρ_v ($\Omega\text{-m}$)	E_{br} (MV/m)
<i>Low frequency dielectrics</i>				
<i>Ceramics</i>				
• electric grade porcelain	6.5	200	10^{11}	25
• ferroelectric ceramics T-7500	7500	300	10^7	2.5
• ferroelectric ceramics VK-1 for variconds	$(20-100) \times 10^3$	1800	10^6	2.5
• piezoelectric ceramics T-1700	1700	300	10^6	2.5
<i>Plastics</i>				
• polyvinyl chloride	4-13	200	10^{12}	16-30
• powdered phenolic plastic with wood dust	11-5	1100-2700	10^5-10^9	10-11
• same with minear filler	10-6	500-1000	$10^{10}-10^{11}$	18
• epoxy plastics	4-2.6	400-500	10^9-10^{12}	16-20

(Contd.)

Table 14.7 (Contd.)

Material	ϵ	$\tan \delta \times 10^4$	$\rho_v (\Omega\text{-m})$	$E_{br} (\text{MV/m})$
<i>High frequency dielectrics</i>				
<i>Ceramics</i>				
• ultraporcelain	8.5	2-10	10^{10}	20
• alumina	9.5	2	10^{13}	15
• steatite	6.5	3-9	10^9	20
• celsian ceramics	7.5	3	10^{10}	35
• ticond T-150	150	3	10^8	10
• thermocond T-20	20	3	10^8	10
<i>Plastics</i>				
• polyethylene	2.4	2-5	10^{13}	20
• polystyrene	2.5	3-5	10^{13}	30
• polytetrafluoroet hylene	2	2	10^{15}	20
<i>Glass</i>	3.5-16	18-175	$10^{11}\text{-}10^{16}$	30-100
<i>Glass ceramics</i>	5-7	3-20	$10^{12}\text{-}10^{16}$	28-48

Spontaneous polarization of this group of materials results as a consequence of interactions between adjacent permanent dipoles where in they mutually align, all in the same direction. For e.g., with BaTiO_3 , the relative displacements of O^{2-} and Ti^{4+} ions are in the same direction for all the unit cells within some volume region of the specimen. At room temperature, ϵ for BaTiO_3 may be as high as 5000. Consequently, capacitors made from these ferroelectric materials can be significantly smaller than capacitors made from other dielectric materials. This is why that these materials are widely used in the manufacture of miniaturized capacitors.

11.9 Electrostriction

This is a mechanical deformation which always accompanies polarization in a dielectric. An electric field polarizes any material by inducing dipole moments. This displacement of charges from their equilibrium positions alters the mechanical dimensions of a solid; it causes electrostriction. However, mechanical stress applied to a neutral material cannot induce dipole moments; i.e., *electrostriction has no inverse*. If a mechanical distortion creates a voltage, the effect must be caused by permanent dipole moments anchored in the structure without a centre of symmetry.

11.10 Piezoelectricity or Pressure Electricity

Electricity, or electric polarity, resulting from the application of mechanical pressure on a dielectric crystal is called piezoelectricity or pressure electricity. The application of a mechanical stress produces in certain dielectric (electrically non-conducting) crystals an electric polarization (electric dipole moment per cubic meter) which is proportional to this stress. If the crystal is isolated, this polarization manifests itself as a voltage across the crystal, and if the crystal is short-circuited, a flow of charge can be observed during loading. Conversely, application of a voltage between certain faces of the crystal produces a mechanical distortion of the material. This reciprocal relationship is referred to as the piezoelectric effect. The phenomenon of generation of a voltage under mechanical stress is referred to as the *direct piezoelectric effect*, and the mechanical strain produced in the crystal under electric stress is called the *converse piezoelectric effect*. The piezoelectric effect is demonstrated in Fig. 14.16.

Piezoelectricity occurs only in insulating materials. Only few ceramic materials exhibit this property. Piezoelectric materials are used extensively in *transducers* for converting a mechanical strain into an electrical signal. Such devices include microphones, phonograph pickups, vibration-sensing elements, and the like. The converse effect, in which a mechanical output is derived from an electrical signal input, is also widely used in such devices as sonic and ultrasonic transducers, headphones, loudspeakers, and cutting

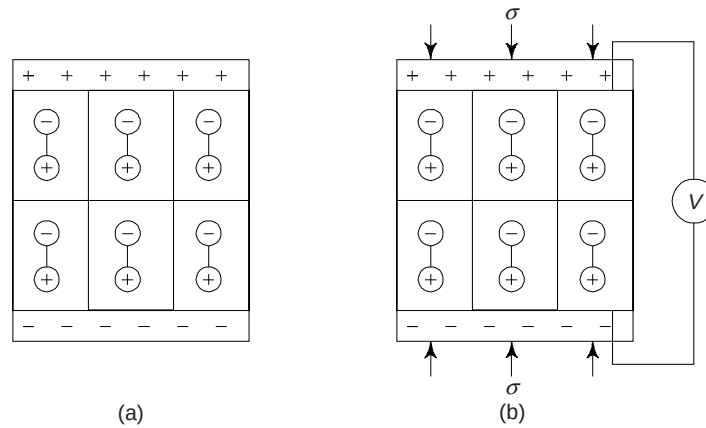


Fig. 14.16 Piezoelectric effect (a) dipoles within a piezoelectric material (b) a voltage is generated when the material is subjected to a compressive stress

heads for disk recording. Both the direct and converse effects are employed in devices in which the mechanical resonance frequency of the crystal is of importance. Such devices include electric wave filters and frequency control elements in electronic oscillator circuits.

The necessary condition for the piezoelectric effect is the absence of symmetry in the crystal structure. Of the 32 crystal classes, 21 lack a centre of symmetry, and with the exception of one class, all of these are piezoelectric. Piezoelectric materials include titanates of barium and lead, zirconate (PbZrO_3), ammonium dihydrogen phosphate ($\text{NH}_4\text{H}_2\text{PO}_4$), natural quartz. As stated earlier, this property is a characteristic of materials having complicated crystal structure with a low degree of symmetry. One may improve the piezoelectric behaviour of a polycrystalline specimen by heating above its Curie temperature and then cooling to room temperature in strong electric field.

Piezoelectrics are required to have high piezoelectric modulus and low losses. Ferroelectric ceramics have the piezoelectric modulus around 6×10^{-10} C/N, which is one or two order of magnitude higher than that of quartz. The structure of piezoelectric ceramics is a solid solution of barium titanate, barium niobate, lead niobate or lead titanate.

Piezoelectric materials have the following important applications: (i) frequency resonators (ii) gramophone pickups (iii) filters (iv) ultrasonic flaw detectors (v) underwater sonar transducers and (vi) air transducers (ear-phones, microphones, hearing aids, etc.).

11.11 Uses of Dielectrics

We have seen that dielectric materials are electrically insulative, yet susceptible to polarization in the presence of an electric field. This polarization phenomenon accounts for the ability of the dielectrics to increase the charge storing capability of capacitors. Now, we can summarize the main uses of dielectrics as follows: (i) piezoelectric and electro-optic devices (ii) in capacitors, resistors and strain gauges (iii) thermionic valves, radiation detectors, electric devices, dielectric amplifier (iv) dielectrics are usually used as ordinary insulators in power cables, signal cables, electric motors, etc. (v) dielectrics are used in transformers and various form of switchgear and generators where the dissipation problem of heat is active, and a common way of getting rid of it is to insulate with a transformer oil, i.e. mineral oil.

Example 4 Show that the relative dielectric constant of a barium titanate crystal, which, when inserted in a parallel plate condenser of area $10 \text{ mm} \times 10 \text{ mm}$ and distance of separation of 2 mm , gives a capacitance of 10^{-9} F is 2259. [M.Sc.(M.S.)]

Solution We have

$$C = 10^{-9} \text{ F}$$

$$d = 2 \times 10^{-3} \text{ m}$$

$$\begin{aligned}
 \epsilon_0 &= 8.854 \times 10^{-12} \\
 A &= 10 \times 10^{-3} \times 10 \times 10^{-3} = 10^{-4} \text{ m}^2 \\
 C &= \frac{\epsilon_r \epsilon_0 A}{d} \\
 \therefore \epsilon_r &= \frac{Cd}{\epsilon_0 A} \\
 &= \frac{10^{-19} \times 2 \times 10^{-3}}{8.854 \times 10^{-12} \times 10^{-4}} \\
 &= 2259
 \end{aligned}$$

Example 5 Using the given data for BaTiO_3 crystal, calculate the polarization. The shift of the titanium ion from the body centre is $0.06 \text{ \AA}$. The oxygen anions of the side faces shift by $0.06 \text{ \AA}$, while the oxygen anions of the top and bottom faces shift by $0.08 \text{ \AA}$, all in a direction opposite to that of the titanium ion. [B.E]

Solution Let us calculate the dipole moments due to the effective number of each type of ion in the unit cell of BaTiO_3 as:

(i) dipole moment due to two O^{2-} ions on the four side faces

$$\mu_1 = Qd = 2 \times 2 \times 1.6 \times 10^{-19} \times 0.06 \times 10^{-10} \text{ C-m}$$

(ii) dipole moment due to one O^{2-} on top and bottom

$$\mu_2 = Qd = 1 \times 2 \times 1.6 \times 10^{-19} \times 0.08 \times 10^{-10} \text{ C-m}$$

(iii) dipole moment due to one Ti^{4+} ion at body centre

$$\mu_3 = Qd = 1 \times 4 \times 1.6 \times 10^{-19} \times 0.06 \times 10^{-10} \text{ C-m}$$

$$\therefore \text{Total dipole moment} = 1.02 \times 10^{-29} \text{ C-m}$$

Now, the polarization is the total dipole moments per unit volume. Now, ignoring the effect due to barium ions, one obtains

$$\begin{aligned}
 P &= \frac{\mu}{A} = \frac{1.02 \times 10^{-29}}{4.03 \times (3.98)^2 \times 10^{-30}} \\
 &= 0.16 \text{ C-m}^2.
 \end{aligned}$$

12. MAGNETISM

This comprises those physical phenomena involving magnetic fields and their effects upon materials. Magnetic fields may be set up on a macroscopic scale by electric currents or by magnets. On atomic scale, individual atoms cause magnetic fields when their electrons have a net magnetic moment as a result of their angular momentum. A magnetic moment arises whenever a charged particle has an angular momentum. It is the cooperative effect of the atomic magnetic moments which causes the macroscopic magnetic field of a permanent magnet. However, the underlying principles and mechanisms that explain the magnetic phenomenon are complex and subtle, and their understanding has eluded physicists until relatively recent times. Several of our modern technological devices rely on magnetism and magnetic materials; these include electric motors, electric power generators and transformers, components of sound and video reproduction systems, telephones, radio, television, computers, etc.

Well known examples of magnetic materials which exhibit magnetic properties are: iron, some steels and the naturally occurring mineral lodestone. The important facts about the magnetic materials are:

- (i) There are some materials which exhibit magnetic properties even without the application of any magnetic field and become more magnetic when a weak magnetic field is applied to them.
- (ii) There are many other materials which lose their initially strong magnetism when heated above a certain critical temperature and become comparatively weakly magnetised.

- (iii) There are some materials which show a magnetic response in a direction opposite to that of any externally applied field.
- (iv) Magnetic materials are all media capable of being magnetized in a magnetic field, i.e. of creating their own magnetic field. According to their magnetic properties, such materials are divided into three principal groups: *diamagnetic*, *paramagnetic* and *ferromagnetic materials*. From the applications point of view all the magnetic materials can be placed under two groups: (i) Soft and (ii) Hard magnetic materials.

Iron, some steels, and the naturally occurring mineral lodestone are well-known examples of materials which exhibit magnetism. Ferro and ferrimagnetic materials are the most important magnetic materials from the point of view of practical applications. We will first consider the terms and definitions used in magnetism.

11.1 Terminology

(i) *Magnetic Induction or Magnetic Flux Density (B)*: In the presence of magnetic field in vacuum, the magnetic induction (B) is related to the field strength H (in units of $A-m^{-1}$) as follows:

$$B = \mu_0 H \quad (30)$$

where μ_0 is called the *permeability of free space*. B is expressed in units of *tesla* or Weber per square metre. The units of permeability are

$$\begin{aligned} \mu_0 &= \frac{B}{H} = \frac{Wb - m^2}{A - m^{-1}} \quad (\because Wb - m^2 = NA^{-1}m^{-1}) \\ &= H - m^{-1} \quad (H = Wb - A^{-1} \text{ where } H \rightarrow \text{Henry}) \end{aligned}$$

In SI units, the permeability of free space (μ_0) has a value of $4\pi \times 10^{-7} H-m^{-1}$.

(ii) *Magnetic Field (H)*: It is said to occupy a region when the magnetic effect of an electric current or of a magnet upon a small test magnet which is brought in the vicinity is detectable. Magnetic field strength is denoted by H . When a magnetic material is placed in a magnetic field H , it becomes magnetized, i.e., it becomes itself a magnet. Magnetic field strength (H) is expressed in units of $A-m^{-1}$.

If the magnetic field is applied to a solid medium, the magnetic induction in the solid is given by a relationship

$$B = \mu H \quad (31)$$

where μ is the permeability of the solid material through which the magnetic lines of force pass. In general μ is not equal to μ_0 . The ratio μ/μ_0 is the relative permeability of the medium and designated by μ_r . Mathematically,

$$\mu_r = \frac{\mu}{\mu_0} \quad (32)$$

(iii) *Magnetization*: This may be defined as the process of converting a non-magnetic bar into a magnetic bar. This term is almost analogous to the polarization in dielectric materials. The flux density

$$\begin{aligned} B &= \mu H = \mu_0 \mu_r H \\ &= \mu_0 \mu_r H + \mu_0 H - \mu_0 H \\ &= \mu_0 H + \mu_0 H (\mu_r - 1) \\ &= \mu_0 H + \mu_0 M \\ &= \mu_0 (H + M) \end{aligned} \quad (33)$$

where $M = \mu_0 (H + M)$ is called the magnetization of solid and expressed in ampere/metre. From the above relation, we find that if a magnetic field is applied to a material, the magnetic flux density is equal to the effect on vacuum and on the material. The magnetization (M) may thus be defined as the magnetic dipole moment per unit volume of the bar.

(iv) *Magnetic Susceptibility (χ)*: The magnitude of the magnetization, M is proportional to the applied field as follows:

$$M = \chi H \quad (35)$$

where χ is called the magnetic susceptibility, which is *unitless*. The magnetic susceptibility and the relative permeability are related as follows:

$$\chi = \mu_r - 1 = M/H \quad (36)$$

We may note that B , M and H are vectors. Magnetic units and conversion factors for SI and CGS – emu systems are given in Table 14.8. We may note that magnetic units may be a source of confusion because there are really two systems in common use. The ones used thus far are SI [rationalized MKS (metre-kilogram-second)]; the others come from the CGS-emu (centimetre-gram-second-electromagnetic unit) system.

Table 14.8 Magnetic units and conversion factors for the SI, CGS and emu systems

Quantity	Symbol	SI units		CGS-emu unit	Conversion
		Derived	Primary		
• Magnetic induction (flux density)	B	tesla (Wb/m ²)	kg/s–C	gauss	$1 \frac{\text{Wb}}{\text{m}^2} = 10^4 \text{ gauss}$
• Magnetic Field Strength	H	$\frac{\text{amp–turn}}{\text{m}}$	C/m–s	oersted	$\frac{1 \text{ amp–turn}}{n} = 4\pi \times 10^{-3} \text{ oersted}$
• Magnetization	M (SI) I (CGS – emu)	$\frac{\text{amp–turn}}{\text{m}}$	C/m–s	maxwell/cm ²	$\frac{1 \text{ amp–turn}}{\text{m}} = 10^{-3} \text{ maxwell/cm}^2$
• Permeability of a vacuum	μ_0	henry/m	kg–m/C ²	unitless	$4\pi \times 10^{-7} \text{ henry/m} = 1 \text{ emu}$
• Relative permeability	μ_r (SI) μ' (CGS-emu)	unitless	unitless	unitless	$\mu_r = \mu'$
• Susceptibility	χ (SI) χ' (CGS-emu)	unitless	unitless	unitless	$\chi = 4\pi\chi'$

Note: units of Weber (Wb) are volt-seconds, units of Henry are Webers per ampere

(v) *Magnetic Dipoles*: Magnetism is *dipolar*, i.e. magnetism is characterized by having two opposite poles: north (N) and south (S). Magnetic dipoles are found to exist in magnetic materials, which, in some respects are analogous to electric dipoles. The strength of a magnetic dipole is measured by the product of the pole strength and the distance between the poles. This is called *magnetic moment*. Magnetic dipoles are influenced by magnetic fields and within a magnetic field, the force of the field itself exerts a torque that tends to orient the dipoles with the magnetic field. One source of magnetism in an atom is the orbital motion of electrons. Each electron revolving around the nucleus in an atom constitutes a circulating electric charge or current and thus produces a small magnetic field. Moreover, each spinning electron on its axis also can be conceived as a circulating charge and also produced a small magnetic fields. Many times it is convenient to think of magnetic forces in terms of fields. Imaginary lines of force may be drawn to indicate the direction of the force at positions in the vicinity of the field source.

11.2 Origins of Magnetic Moments

The macroscopic magnetic properties of a substance are a consequence of *magnetic moments* associated with individual electrons. Each electron in an atom has magnetic moments that originate from the following two sources:

- (i) orbital magnetic moment of electrons
- (ii) spin magnetic moment of electrons.

We may note that permanent magnetic moments can also arise from *spin magnetic moment of the nucleus*. Of the three, spin dipole moments of electrons are important in most magnetic materials. Magnetic moments associated with (a) an orbiting electron and (b) a spinning electron is shown in Fig. 14.17. Electron in an atom is continuously orbiting around the nucleus; being a moving charge, an electron may be considered to be a small current loop, generating a very small magnetic field, and having a magnetic moment along its axis of rotation (Fig. 14.17(a)). Moreover, each electron may also be thought of as spinning around an axis; the other magnetic moment originates from this electron spin which is directed along the spin axis as shown in Fig. 14.17(b). We may note that spin magnetic moments may be only in an “up” direction or in an antiparallel “down” direction. Obviously, each electron in an atom may be thought of as being a small magnet having permanent orbital and spin magnetic moments. The net magnetic moment due to electron spin in a sodium atom is one unit, called a *Bohr magneton* μ_B , which is of magnitude $9.27 \times 10^{-24} \text{ A-m}^2$. For each electron in an atom the spin magnetic moment is $\pm \mu_B$ (+sign for spin up and–sign for spin down). The orbital magnetic moment contribution is equal to $m_l \mu_B$, m_l being the magnetic quantum number of the electron.

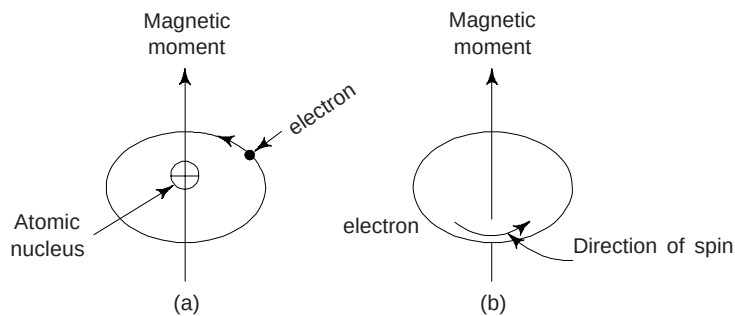


Fig. 14.17 Magnetic moment associated with (a) an orbiting electron and (b) a spinning electron

Magnetic properties of a material originate due to the imbalance of spin orientation in atoms. A number of atoms or molecules have paired and unpaired electrons. We may note that there is one electron in the outermost s-shell which is unpaired and align itself in an applied field, which gives rise to magnetism. This reveals that the magnetic moment of an atom in the solid state is due only to an incomplete inner shell. It is interesting to note that the electrons in the incomplete inner shells of transition elements have their spins aligned in the same direction, or one may say that the electrons arrange themselves among the energy levels to give the maximum possible total spin angular momentum. This is quite consistent with Pauli exclusion principle. Thus the net magnetic moment, then for an atom is just the sum of the magnetic moments of each of the constituent electrons, including both orbital and spin contributions, and taking into account moment cancellation. For an atom having completely filled electron shells or subshells, when all electrons are considered, there is total cancellation of both orbital and spin moments. This is why the materials composed of atoms having completely filled electron shells are not capable of being permanently magnetized. This type of materials include the inert gases. (He, Ne, Ar, etc.) as well as some ionic materials.

One can obtain the atomic moments of 3rd transition elements in the solid state by adding the spin magnetic moments of the electrons in the unfilled shell.

11.3 Classification of Magnetic Materials

There are three classes into which all the magnetic materials may be grouped according to their magnetic behaviour, although there is some overlap among groups:

1. Diamagnetic substances (Fig. 14.18a)

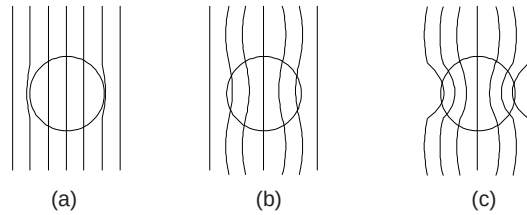


Fig. 14.18 (a) Diamagnetic solid: Lines of forces are slightly repulsed. Relative permeability is less than 1 (b) Paramagnetic solid: Lines of forces are attracted weakly. Relative permeability is slightly greater than 1. (c) Ferromagnetic solid: Lines of forces are attracted very strongly. Relative permeability is much greater than 1

2. Paramagnetic substances (Fig. 14.18b)

3. Ferromagnetic substances (Fig. 14.18c)

In addition, *antiferromagnetism* and *ferrimagnetism* are considered to be the subclasses of ferromagnetism.

(i) *Diamagnetism*: Diamagnetism is a very weak form of magnetism exhibited by substances with a negative magnetic susceptibility ($\chi = \mu_0 M/B$), i.e., by substances which magnetize in a direction opposite to that of an applied magnetic field. A diamagnetic substance has a magnetic permeability less than 1, and is repelled when placed near a magnet. The examples are organic solids like naphthalene, benzene, etc.; metals like silver, gold and copper; atoms with rare gas configurations like, A, He, Ne, etc. The magnetization of diamagnetic substances is associated with the currents induced on application of a magnetic field. According to Lenz's law, the flow of an induced current is in such a direction as to oppose the change of flux of inducing field; this accounts for the negative susceptibility. The diamagnetic susceptibility is invariably small, of the order of $10^{-5} \text{ cm}^3/\text{mole}$. When placed between the poles of a strong electromagnet, diamagnetic materials are attracted toward regions where the field is weak.

The atomic magnetic dipole configuration for a diamagnetic material with and without an external field is shown in Fig. 14.19. The arrows in the figure represent atomic dipole moments. Figure 14.20 shows the dependence of B on the external field H for a material which exhibits diamagnetic behaviour.

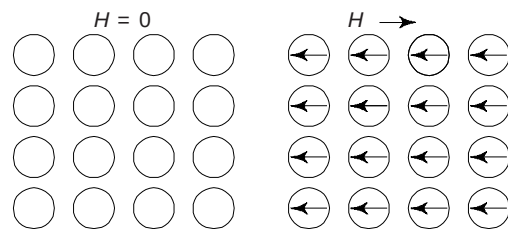


Fig. 14.19 Schematic illustration of the atomic dipole configuration for a diamagnetic material with and without a magnetic field. No dipoles exist in the absence of an external field, whereas in the presence of a field, dipoles are induced that are aligned opposite to the field direction

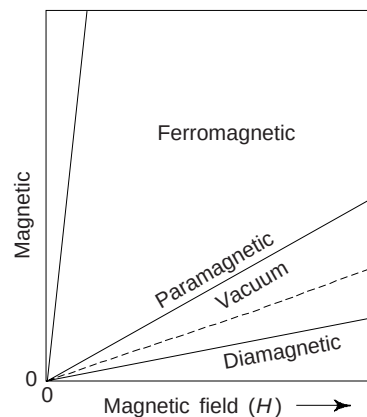


Fig. 14.20

Table 14.9 Magnetic susceptibilities for diamagnetic materials at room temperature

Material	Susceptibility, χ (volume) (SI units)
Copper	-0.96×10^{-5}
Gold	-3.44×10^{-5}
Mercury	-2.85×10^{-5}
Silicon	-0.41×10^{-5}
Silver	-2.38×10^{-5}
Zinc	-1.56×10^{-5}
Aluminium oxide	-1.81×10^{-5}
Sodium Chloride	-1.41×10^{-5}

Diamagnetism is found in all materials; but because it is so weak, it can be observed only when other types of magnetism, i.e. para or ferromagnetism are totally absent. The condition for pure diamagnetism is that all electronic spins be paired and all orbital moments either be zero or effectively cancel one another. Nearly all molecules with an even number of electrons satisfy this condition; an important exception is O_2 . The condition is also satisfied by most non-metallic solids, except compounds containing atoms with incomplete innershell electron groups, such as the transition rare-earth, and actinide elements.

The diamagnetic response of a substance is small; only a very small fraction of the applied magnetic field is shielded from the interior of the substance by the induced diamagnetic currents. There is one case, however, in which the inducing field is completely shielded (except for small surface effects). This is the perfect diamagnetism exhibited by superconductors and is known as Meissner effect (see chapter 16). We may note that diamagnetism is of no practical importance.

(ii) *Paramagnetism*: A property exhibited by substances which, when placed in a magnetic field, are magnetized parallel to the field to an extent proportional to the field (except at very low temperatures or in extremely large magnetic fields). Paramagnetic materials always have permeabilities greater than 1, but the values are in general not nearly so great as those of ferromagnetic materials.

For some solid materials, each atom possesses a permanent dipole moment by virtue of incomplete cancellation of electron spin and/or orbital magnetic moments. In the absence of an external magnetic field, the orientations of these atomic magnetic moments are random, such that a piece of material possesses no net macroscopic magnetization. These atomic dipoles are free to rotate, and *paramagnetism* results when they preferentially align, by rotation, with an external field as illustrated in Fig. 14.21. These magnetic dipoles are acted on individually with no mutual interaction between adjacent dipoles. In as much as the dipole align with the external field, they enhance it, giving rise to a relative permeability μ_r that is greater than 1, and to a relatively small and positive susceptibility. Susceptibilities for few paramagnetic substances are given in Table 14.10. Susceptibilities for paramagnetic substances range from about 10^{-5} to 10^{-2} . A schematic B - H curve for a paramagnetic substance is also shown in Fig. 14.20.

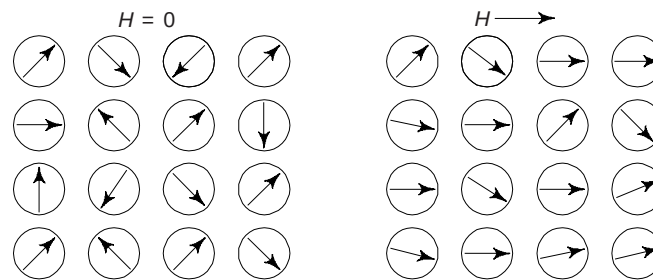
**Fig. 14.21 Atomic dipole configuration with and without an external magnetic field for a paramagnetic material**

Table 14.10 Magnetic susceptibilities for paramagnetic materials at room temperature

Materials	Susceptibility χ (volume) (SI units)
Aluminium	2.07×10^{-5}
Chromium	3.13×10^{-4}
Chromium chloride	1.51×10^{-3}
Manganese sulphate	3.70×10^{-3}
Molybdenum	1.19×10^{-4}
Sodium	8.48×10^{-6}
Titanium	1.81×10^{-4}
Zirconium	1.09×10^{-4}

The following types of substances are paramagnetic:

- All atoms and molecules which have an odd number of electrons. According to quantum mechanics, such a system cannot have a total spin equal to zero; therefore, each atom or molecule has a net magnetic moment which arises from the electron spin angular momentum. Examples are organic free radicals and gaseous nitric oxide.
- All free atoms and ions with unfilled inner electron shells and many of these ions when in solids or in solution. Examples are transition, rare-earth and actinide elements and many of their salts. This includes ferromagnetic and antiferromagnetic materials above their transition temperatures.
- Several miscellaneous compounds including molecular oxygen and organic biradicals.
- Metals, in this case, the paramagnetism arises from the magnetic moments associated with the spins of the conduction electrons and is called the *Pauli paramagnetism*.

Relatively few substances are paramagnetic. Aside from the Pauli paramagnetism found in metals, the most important paramagnetic effects are found in the compounds of the transition and rare-earth elements which have practically filled $3d$ and $4f$ electron shells respectively.

Most paramagnetic substances at room temperature have a static susceptibility which follows a Langevin-Deby law (Eq. (37)),

$$\chi = \frac{Np^2 \mu_B^2}{3kT + N\alpha} \quad (37)$$

where N is the number of magnetic dipoles per unit volume, p is the effective magneton dipoles per μ_B is the Bohr magneton, k is Boltzmann's constant, T is absolute temperature and α is the temperature independent contribution of Van-Vleck paramagnetism.

Both paramagnetic and diamagnetic substances are considered to be non magnetic because they exhibit magnetization only in the presence of an external field. Moreover, for both, the flux density B within them is almost the same as it would be in vacuum.

(iii) *Ferromagnetism*: A property exhibited by certain metals, alloys and compounds of the transition (iron groups as BCC α ferrite and cobalt and nickel), rare earth metals such as gadolinium (Gd), and actinide elements in which, below a certain temperature called the *Curie temperature* (Eq. 38),

$$\chi = \frac{C}{T - \theta} \quad (38)$$

We must note that the general behaviour of the susceptibility of ferromagnetic materials above the Curie temperature, T_c , follows the Curie-Weiss law (Eq. (38)). The behaviour is followed in the region well above the ferromagnetic curie temperature T_c . The paramagnetic curie temperature θ is usually slightly greater than the temperature of transition T_c . Comparison of T_c and θ for three ferromagnetic metal is given below comparison of θ and T_c .

Comparison of θ and T_c

Parameter	Fe	Co	Ni
θ (K)	1093	1428	650
T_c (K)	1043	1393	631

In a region just a fraction of degree above the “critical point”, or Curie temperature T_c , the susceptibility is found to approximate the following relation,

$$\chi = \frac{C'}{(T - T_c)^\gamma}$$

with γ generally very close to 1.33. We may note that the theory is extremely complicated and not entirely satisfactory], the atomic magnetic moments tend to line up in a common direction. All ferromagnetic materials exhibit paramagnetic behaviour above their ferromagnetic Curie point. The Curie temperature makes a transition between order and disorder of the alignment of the atomic magnetic moments. Some materials exhibit a special form of ferromagnetism below the Curie temperature called *ferrimagnetism*. The magnitude of the paramagnetic susceptibility is determined by the Curie constant C (Eq. 38). A typical value of C is 0.2 K/cm^3 for iron. Below the Curie point, the static susceptibility is not usually defined for a ferromagnetic substance, since the ferromagnet may have a finite magnetization in zero applied field. Magnetic susceptibilities as high as 10^6 are possible for ferromagnetic substances. Consequently, $H \ll M$, and from Eq. (34), we write

$$B \cong \mu_o M \quad (39)$$

Some physical properties of ferromagnetic elements are summarized in Table 14.10.

Table 14.10 Some selected physical properties of ferromagnetic elements

Element	Electronic configuration	Crystal structure	Magnetization at 0 K (Amp/m)	Ferromagnetic curie temp. T_c (K)	Melting temp. (K)
Fe	$3d^6 4s^2$	bcc	1.7×10^6	1043	1810
Co	$3d^7 4s^2$	hcp	1.4×10^6	1404	1750
Ni	$3d^8 4s^2$	fcc	0.48×10^6	632	1732
Gd	$4f^7 5d^1 6s^2$	hcp	5.66×10^6	290	1586

Permanent magnetic moments in ferromagnetic materials result from atomic magnetic moments due to electron-spin-uncancelled electron spins as a consequence of the electron structure. There is also an orbital magnetic moment contribution that is small in comparison to the spin moment. Moreover, in a ferromagnetic material, coupling interactions cause net spin magnetic moments of adjacent atoms to align with one another, even in the absence of an external field (Fig. 14.22).

The characteristic property of a ferromagnet is that, below the curie temperature, it can possess a spontaneous magnetization in the absence of an applied magnetic field. Upon application of a weak magnetic field, the magnetization increases rapidly to a high value called the *saturation* magnetization, which is a general function of temperature (Eq. 40),

$$\frac{M}{M_{\text{sat}}} = \frac{T\alpha}{\theta} \quad (40)$$

where M_{sat} is saturation value of magnetization, θ is the paramagnetic Curie temperature and $\alpha = \mu_o \mu_B \gamma M / k_B T$,

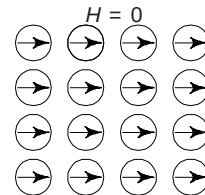


Fig. 14.22 Mutual alignment of atomic dipoles for a ferromagnetic material, which exist even in the absence of external field ($H = 0$)

where γM is the Weiss field. Obviously, for a given temperature T , Eq. (40) is a plot of M/M_s versus α represent a straight line with a slope equal to T/θ . For $T \geq \theta$, there is no spontaneous magnetization.

The task of a successful theory of ferromagnetism are to account for the spontaneous magnetism below the Curie point, the temperature dependence of the saturation magnetization, and the nature of the magnetization process, or magnetization curve.

Weiss Theory

The Weiss molecular field theory of ferromagnetism represents the first realistic attempt to account for the properties of a ferromagnet. This theory rests on two hypotheses:

- (i) Below the Curie point, a ferromagnetic substance is composed of small, spontaneously magnetized regions called *domains*. The total magnetic moment of the material is the vector sum of the magnetic moments of the individual domains. It is now known that these assumed domains really exist and are usually 0.1 and 0.01 cm across.
- (ii) Each domain is spontaneously magnetized because a strong molecular (magnetic) field tends to align the individual atomic magnetic moments within the domain.

The consequence of these assumptions is that, while each domain is spontaneously magnetized, the directions of magnetization of the domains do not coincide; therefore the overall magnetization of the sample may be much smaller than if it were composed of a single domain. Application of a relatively weak field (~100-1000 Oersteds) and (often very much less) is sufficient to align the directions of the magnetization of the domains, thereby achieving a large magnetization.

The second hypothesis of the Weiss theory leads to the existence of a Curie temperature below which a domain may be spontaneously magnetized in the absence of an applied magnetized field. According to the theory, the effective field acting on any atomic magnetic moment within the domain may be written as,

$$H = H_o + \lambda M \quad (42)$$

where H_o is an externally applied magnetic field and λM (M = Magnetization) is the Weiss molecular field whose order of magnitude in Fe is 10^7 Oersteds.

There is an overall general agreement between theory and experimental results of temperature dependence of spontaneous magnetization for Fe, Ni and Co. The low temperature behaviour, however, is better described by *magnon theory*. The theory satisfactorily describes the temperature dependence of susceptibility in the paramagnetic region provided the temperature is well above the Curie temperature. The Curie temperature determined from the theory of spontaneous magnetization differs by a few degrees from the experimentally determined value for the paramagnetic region.

Heisenberg Exchange Interaction Theory

The Heisenberg theory of ferromagnetism treats the origin of the Weiss molecular field on an atomic basis. It may be remarked that the ordinary dipole-dipole interactions among atomic magnetic moments are much too small to account for the Weiss field. The foundation of the Heisenberg theory is the Pauli exclusion principle. Due to Heisenberg exchange interaction two neighbouring spins in a solid are coupled together with an exchange energy given by

$$E_{\text{exch}} = -2 J \vec{S}_i \cdot \vec{S}_j \quad (42)$$

where $\vec{S}_i$ and $\vec{S}_j$ are the spin angular momentum vectors of the two electrons i and j , and J is the so called exchange integral between the two electrons. The exchange integral decreases rapidly with distance between the electrons and depends in a complicated way upon the spatial distribution (wave function) of the electrons. It is extremely difficult to compute. The strength of J can be estimated from the Curie temperature T_c . If there are Z nearest neighbours to a central i^{th} spin, the exchange energy for this spin is

$$E_{\text{exch}} = -2 \sum_{j=1}^Z J_{ij} \vec{S}_i \cdot \vec{S}_j \approx -2 ZJS^2 \quad (43)$$

where we have assumed that all the J_{ij} are equal to J and the spins are parallel. This exchange energy must be equal to kT_c since at T_c the magnetic order is destroyed. Thus

$$J \approx \frac{k_B T_c}{2ZS^2} \quad (44)$$

If the exchange integral is positive, the parallel arrangement is favoured, and if J is large enough, ferromagnetism should result. If J is large and negative, antiferromagnetism or ferrimagnetism supposedly arises. The order of magnitude of J given by $J \sim kT_c \sim 10^{-20}$ Joule.

There seems to be no question that the Heisenberg theory correctly accounts for the tendency of the electrons in the same ferromagnetic atom to exhibit parallel spins. However, whether or not, it leads to the correct explanation of the interatomic alignment of spins is still a subject of much controversy. In insulators it usually proves possible to express the coupling between atomic spins by Eq. (42), provided J is interpreted as an effective exchange integral. In metals the problem is much more complicated. There is little doubt, however that the basic Heisenberg idea is correct and that the molecular field arises from the interplay between electrical forces and the effects of Pauli exclusion principle.

Crystalline Anisotropy Energy

This accounts for the experimental fact that ferromagnets tend to magnetize along certain crystallographic axis, called directions of easy magnetization. For example, a single crystal of Fe, which is made up of a cubic array of iron atoms, tends to magnetize in the direction of the cube edges. It requires about 1.4×10^4 J/m³ (at room temperature) to move the magnetization into a hard direction along a cubic body diagonal.

The Heisenberg exchange energy, Eq. (42) is isotropic and cannot account for the observed anisotropy, which probably has its origin in a complicated interplay of spin-orbit coupling, crystalline electric fields, and overlap of orbital wave functions. Anisotropy energy depends on the strain of the crystal, giving rise to *magnetostriction*, i.e., changes in length of a substance when it is magnetized.

Ferromagnetic Domains

Small regions of spontaneous magnetization, formed at temperatures below the Curie point, are known as domains. Domains originate in order to lower the magnetic energy as illustrate in Fig. 14.23. In Fig. 14.23(b) it is shown that two domains will reduce the extent of the external magnetic field, since the magnetic lines of force are shortened. On further subdivision (Fig. 23(c)), this filled is still further reduced.

Another way to describe the energy reduction is to note that the interior demagnetizing fields, coming from surface poles, are much smaller in the long, thin domain of Fig. 14.23(c) than in the “fat” domain of Fig. 14.23(a).

The question arises as to how long this subdivision process continues. With each subdivision there is a decrease in field energy, but there is also an increase in Heisenberg exchange energy, since more and more

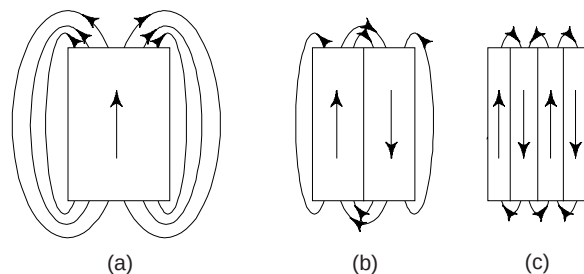


Fig. 14.23 Lowering of magnetic field energy by domains. (a) Lines of force for a single domain. (b) Shortening of lines of force by division into two domains. (c) Reduction of field energy by further subdivision

magnetic moments are aligning antiparallel. Finally a state is reached in which further subdivision would cause a greater increase in exchange energy than it would cause decrease in field energy, and the ferromagnet would assume this state of minimum total energy.

Bloch Wall

Also because of exchange energy, the reversal of magnetization between domains does not occur abruptly but takes place gradually through a transition zone called the Bloch Wall (Fig. 14.24). In iron the wall is ~ 50 nanometers thick and has the total energy $\sim 2 \times 10^{-3} \text{ J/m}^2$.

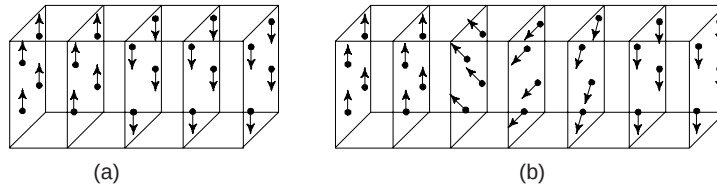


Fig. 14.24 Lowering of exchange energy by the transition zone known as Bloch wall. The reversal of magnetization between domains does not take place abruptly as is shown but by degrees as is illustrated in Fig. 24(b)

Domain Model

The orientation of domains in a crystal is primarily determined by the need to minimize the magnetic energy (Fig. 14.22). It is possible to eliminate all surface magnetic poles by forming flux closure domains (Fig. 14.25). Here the normal component of magnetization is continuous across all domain boundaries. The demagnetizing fields are zero everywhere, except for a trivial effect of surface poles in the Bloch Walls. In a uniaxial crystal, that is, a crystal with a single easy direction, an arrangement as shown in Fig. 14.25(b) will be preferred since it has a lower density of magnetization normal to easy direction, or in the hard direction. Even in cubic crystals, in which all directions of magnetization in Fig. 14.25 may be easy, Fig. 14.25(b) will be preferred because of magnetostriction. In iron, for example, each domain increases in length along the direction of magnetization by a fraction $\sim 2 \times 10^{-5}$. Thus the domains of Fig. 14.25 can be fitted smoothly together only by straining them elastically against this magnetostriction, and the required strain energy will be smaller in Fig. 14.25(b) than in Fig. 14.25(a).

If an external magnetic field is applied to a ferromagnetic specimen, two things can happen, according to the domain model: (i) the domains parallel or nearly parallel to $\vec{H}$ can grow in size at the expense of the antiparallel domains shown in Fig. 14.26, or (b) the magnetic moment of the domains can rotate into the field direction. In either case the sample acquires a magnetization, which increase as the field increase, until all the domains are parallel to the applied field, at which point the material is said to be saturated and the magnetization is equal to $\vec{M}_s$.

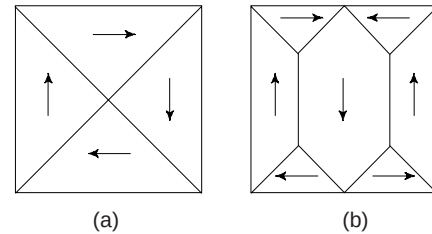


Fig. 14.25 Flux closure domains (single crystal). (a) Large domains at right angles. (b) Reduction in their size, causing reduction of anisotropy energy of uniaxial crystals or of strain energy of cubic crystals

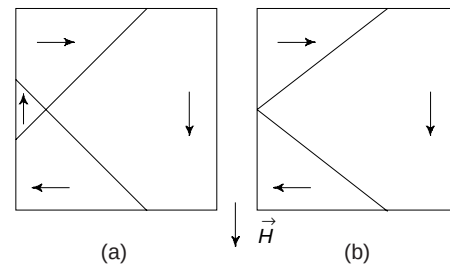


Fig. 14.26 Growth of domains parallel to an applied magnetic field $\vec{H}$

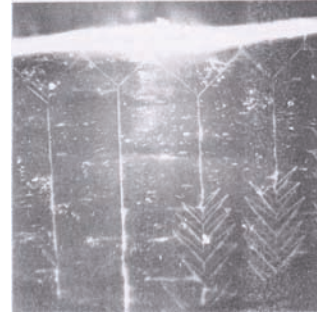
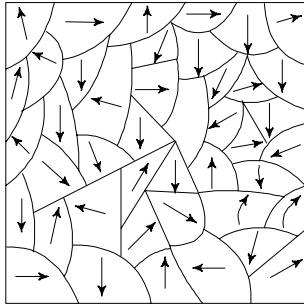


Fig. 14.27 Orientation of domains in a polycrystal **Fig. 14.28** Bitter powder patterns on a (100) surface of silicon-iron

In polycrystals the domain structure is more complicated (Fig. 14.27), depending upon such variables as grain orientation and grain boundaries. It is possible, however, for domains to cross grain boundaries.

On minimizing the total contribution from (i) magnetic, or demagnetizing energy (ii) anisotropy (iii) magnetostriction (iv) elastic strain, and (v) Bloch wall energy, it is found that, depending upon the composition and shape of the crystal, the theoretical domain thickness should vary from about 0.1 to 0.001 cm.

Direct experimental evidence of the existence of domains is furnished by Bitter powder patterns (Fig. 14.28). It is also possible to observe domains by means of transmission or reflection of electrons in an electron microscope. The Kerr magnetooptic effect (rotation of plane of polarization of light reflected from a magnetic surface) has also been used to study domains. These two methods are particularly useful if the Bloch Walls are extremely thick (because of small magneto crystalline anisotropy) and the resultant Bitter powder patterns very blurred.

(iv) *Antiferromagnetism*: This originates when the spin moments of the neighbouring atoms are ordered in an antiparallel arrangement (Fig. 14.29) or when the exchange integral is negative. This property is exhibited by some metals, alloys and salts of transition elements in which the atomic magnetic moments, at sufficiently low temperatures, form an ordered array which alternates or spirals so as to give no net total moment in zero applied field. The most direct way of detecting such arrangements is by means of neutron diffraction. A crystal exhibiting antiferromagnetism may be considered to be consisting of two independent sublattices A and B, one of which is spontaneously magnetized in the opposite direction. This type of magnetism was first observed in the crystals of MnO. Manganese oxide (MnO) is a ceramic material that is ionic in character having both Mn^{2+} and O^{2-} ions. No net magnetic moment is associated with the O^{2-} ions, since there is a total cancellation of both spin and orbital moments. However, the Mn^{2+} ions possess a net magnetic moment that is predominantly of spin origin. These Mn^{2+} ions are arrayed in the crystal structure of MnO crystal such that the moments of adjacent Mn^{2+} ions are antiparallel (Fig. 14.29). This is why that the opposing magnetic moments cancel one another, and as a consequence, the solid as a whole possess no net magnetic moment.

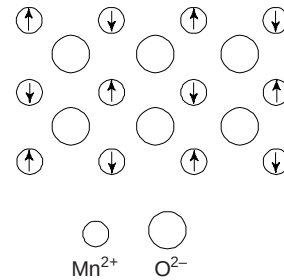


Fig. 14.29 The antiferromagnetism in manganese oxide

When a magnetic field is applied, a small magnetization appears in the direction of the field which increases further with temperature. Such a behaviour is typical of an antiferromagnetic material. The magnetization becomes the maximum at a critical temperature T_N , called the *Neel temperature*, which is analogous to the Curie temperature in the paramagnetic or ferromagnetic substances. This is the transition temperature below which the spontaneous magnetic ordering takes place. Above this temperature, the

magnetization, the magnetization decreases continuously which is indicative of the paramagnetic state of the material. The variation of susceptibility with temperature is governed by

$$\chi = C/(T + \theta) \quad (45)$$

where $\theta = T_N$ is called the Neel temperature. Some representative antiferromagnets are given in Table 14.11.

Table 14.11 Some representative antiferromagnets

Substance	Crystal type	T_N (K)	Paramagnetic curie temperature θ (K)
MnF ₂	Rutile	67	80
MnO	NaCl	122	610
FeO	NaCl	198	507
KMnF ₃	Perovskite	88	158
CuCl ₂ · 2H ₂ O	Orthorhombic	4.3	4.5
CrSb	NiAs	723	550
Cr ₂ O ₃	Al ₂ O ₃	307	485
ZnFe ₂ O ₄	Spinel	9	—
EuTe	NaCl	7.8	6
MnTe	Hexagonal close-packed	403	690

(v) *Ferrimagnetism*: In some compounds, e.g. in some ceramics, etc., the constituent atoms may be antiferromagnetically coupled but with different magnetic moments. This would give rise to a net magnetic moment in each coupling and the some of the moments of all the coupling could result in magnetization which is comparable in order of magnitude to ferromagnets. This phenomenon is termed as ferrimagnetism. The distinction between the two lies in the source of the net magnetic moments. This type of magnetism occurs in materials such as ferrites which are basically the oxides of various metal elements. These ionic materials may be represented by the chemical formula MFe_2O_4 , in which M represents any one of several metallic elements. The prototype ferrite is Fe_3O_4 or $FeO \cdot Fe_2O_3$, the mineral magnetite, sometimes also called as lodestone.

The formula for Fe_3O_4 may be written as $Fe^{2+} O^{2-} - (Fe^{3+})_2 (O^{2-})_3$ in which the Fe ions exist in both +2 and +3 valence states in the ratio of 1 : 2. A net spin magnetic moment exists for each Fe^{2+} and Fe^{3+} ion, which corresponds to $4\mu_B$ and $5\mu_B$ respectively for the two ion types (Fig. 14.30). Moreover, O^{2-} ions are magnetically neutral. We may note that there are antiparallel spin-coupling interactions between the Fe ions, similar in character to antiferromagnetism. However, the net ferrimagnetic moment arises from the incomplete cancellation of spin moments. Net magnetic moments for six cations are given in Table 14.12.

Table 14.12 Net magnetic moments for cations

Cation	Net magnetic moment(μ_B)
Fe ³⁺	5
Fe ²⁺	4
Mn ²⁺	5
Co ²⁺	3
Ni ²⁺	2
Cu ²⁺	1

One can also produce cubic ferrites having other compositions by adding metallic ions that substitute for some of the iron in the crystal structure. Again from the ferrite chemical formula, $M^{2+}O^{2-} - (Fe^{3+})_2 (O^{2-})_3$, in addition to Fe^{2+} , M^{2+} may represent divalent ions such as Ni^{2+} , Mn^{2+} , Co^{2+} and Cu^{2+} , each of which possesses a net spin magnetic moment different from $4\mu_B$ (Table 14.12). Obviously, by adjustment

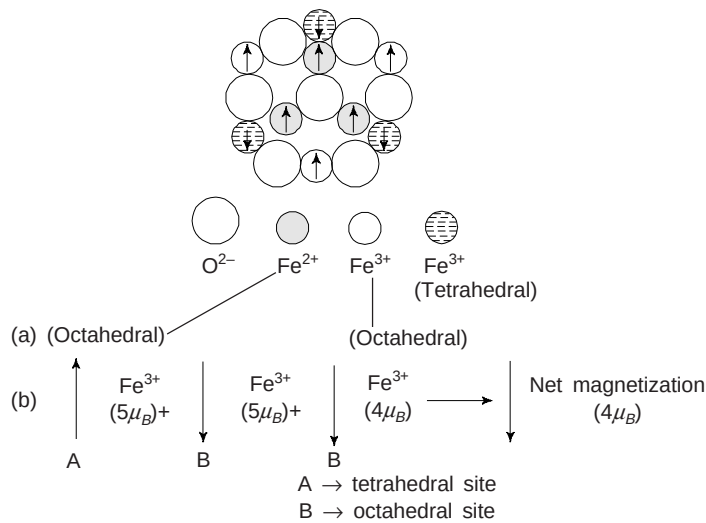


Fig. 14.30 (a) The spin magnetic moment configuration for Fe^{2+} and Fe^{3+} ions in Fe_3O_4 . (b) Magnetizations of the individual sites and the net magnetization in one formula unit of Fe_3O_4

of composition, ferrite compounds having a range of magnetic properties may be produced, e.g., nickel ferrite has the formula NiFe_2O_4 . Mixed ferrites may also be produced containing mixtures of two divalent metal ions such as $(\text{Mn}, \text{Mg})\text{Fe}_2\text{O}_4$ in which $\text{Mn}^{2+} : \text{Mg}^{2+}$ ratio may be varied.

Other than cubic ferrites, ceramic materials are also ferrimagnetic. These include the hexagonal ferrites and garnets. Hexagonal ferrites have a crystal structure similar to the inverse spinel, with hexagonal symmetry rather than cubic. We may write the chemical formula for these materials as $\text{AB}_{12}\text{O}_{19}$, in which A is divalent metal such as Ba, Pb or Sr, and B is a trivalent metal such as Al, Fe, chromium and gallium. $\text{PbFe}_{12}\text{O}_{19}$ and $\text{BaFe}_{12}\text{O}_{19}$ are two most common examples of hexagonal ferrites.

The most versatile of ferrimagnetic systems are the rare earth iron garnets. The garnet unit cell has three sets of inequivalent magnetic sites, differing in their coordination to neighbouring oxygen ions. Obviously, garnets have very complicated crystal structure, which may be represented by the general formula $\text{M}_3\text{Fe}_5\text{O}_{12}$; where M represents a rare earth ion such as samarium, europium, gadolinium or yttrium. Rare-earth ions of various species have widely different magnetic moments, exchange interactions, and crystalline anisotropies. By a suitable choice of rare-earth ions, it is possible to design ferrimagnetic systems with prescribed magnetization and temperature behaviour. Yttrium iron garnet ($\text{Y}_3\text{Fe}_5\text{O}_{12}$), sometimes denoted by YIG, is the most common garnet material.

Ferromagnetic materials which finds wide applications in electrical engineering have a disadvantage, that they have low electrical resistivity. The laminations used for electrical machines have a resistivity $\sim 14 \times 10^{-4} \Omega\text{-m}$ whereas the highest value obtainable in ferromagnetic alloys is less than $10^{-2} \Omega\text{-m}$. Obviously, this disadvantage of the ferromagnetic materials limit their application in high frequency alternating current applications, high eddy current losses and poor magnetic utilization of metals occur in sheets even at low frequencies. The saturation magnetization for ferrimagnetic materials are not as high as for ferromagnets. Moreover, ferrites being ceramic materials, are good electrical insulators. For some magnetic applications, such as high-frequency transformers, a low electrical conductivity is most desirable, because this prevents eddy currents in cores of coils.

11.4 Magnetostriction

The change of length of a ferromagnetic material when it is magnetized is called magnetostriction. More generally, it is the phenomenon that the state of strain of a ferromagnetic sample depends on the direction and extent of magnetization. The phenomenon has an important application in magnetostriction transducers.

Magnetostriction results from the dependence of the crystalline anisotropy energy upon the state of strain of the crystalline lattice. If the crystal deforms (for e.g., suffers a change in length), the anisotropy energy may be lowered more than the elastic energy is raised. Thus a strained state will be favoured.

The total energy of a ferromagnetic substance depends upon the state of strain and the direction of magnetization through three contributions. The first two consist of the crystalline anisotropy energy of the unstrained lattice plus a correction which takes into account the dependence of the anisotropy energy on the state of the strain. The third contribution is that of the elastic energy, which is independent of magnetization direction and is a minimum in the unstrained state. The state of strain of the crystal will be that which makes the sum of the three contributions to the energy a minimum. The result is that, when magnetized, the lattice is always distorted from the unstrained state, unless there is no anisotropy.

Since spontaneous magnetization occurs below the curie temperature (T_c), there will always be a spontaneous lattice distortion which depends on magnetization direction in the ferromagnetic state. In Ni, the lattice spacing parallel to the magnetization is always smaller than the lattice spacing perpendicular to the magnetization.

The magnetostriction coefficient (λ) is defined as the fractional change (extension or contraction) in length, $\Delta l/l$ associated with the change in magnetization from 0 to saturation. The saturation magnetostriction is denoted by λ_s . All domains are oriented in the direction of magnetization, when the material is saturated. In this situation, the expansion in the direction of magnetization is the same at saturation for a single domain or for a number of domains. λ_s is positive in some materials and negative in others. A negative value of λ_s shows that there is contraction in the direction of magnetization and vice versa. We may note that any elastic deformation, whether expansion or contraction, causes an increase in strain energy.

Figure 14.31 shows the magnetostriction versus applied field curves for Ni, Co, Fe and 45 permalloy. From figure it is clear that the dimensions tend to increase in the direction of H and $\perp$ to H for 45 permalloy and Fe whereas for Co and Ni the opposite is true. Ni has the highest negative magnitude and permalloy (45 Ni) has one of the highest positive magnetostrictions.

When a high permeability (soft) magnetic material is required, the magnetostriction should be small in order that anisotropy not be induced by lattice distortions. Magnetostriction appears to be a major source of transformer hum. As the silicon content of soft magnetic steel is increased toward 6.5%, the magnetostriction disappears. Unfortunately, the metal becomes the excessively brittle.

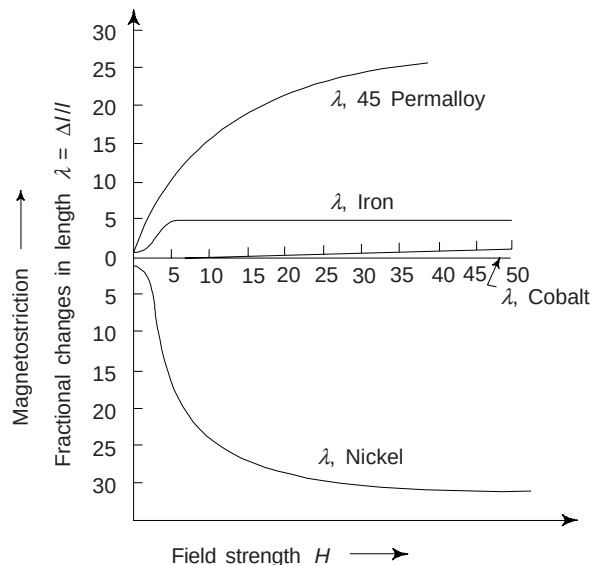


Fig. 14.31 Magnetostriction of some common materials exhibiting expansion and contraction

Magnetostriction is an important property. This effect is exploited in transducers used for the reception and transmission of high frequency sound vibrations. Ni is often used for this application.

Magnetostriction or mechanical deformation of magnetic domains has also other important effects. When a ferromagnetic material is strained, we find that the domains tend to realign themselves into positions of lower energy. Because of this the permeability of the material is changed and it becomes easier or more difficult to magnetize. Realignment puts the domains in a better position to be polarized and permeability is increased in materials having positive magnetostriction, the opposite is true for materials with negative mechanical deformation.

We may note that due to thermal agitation magnetostriction is decreased, because at higher temperatures, the domains are hindered in their reorientation.

11.5 The Influence of Temperature on Magnetic Behaviour

Magnetic characteristics of materials can also be influenced by temperature. We have read that raising the temperature of a solid results in an increase in the magnitude of the thermal vibrations of atoms. Since the atomic magnetic moments are free to rotate and hence with rise in temperature, the increased thermal motion of the atoms tends to randomize the directions of any moments that may be aligned.

For ferromagnetic, antiferromagnetic, and ferrimagnetic materials, the atomic thermal motions counteract the coupling forces between the adjacent atomic dipole moments, thereby causing some dipole misalignment, regardless of whether an external magnetic field is present. This results in a decrease in the saturation magnetization for both ferro- and ferrimagnets. The saturation magnetization is a maximum at 0 K, at which temperature the thermal vibrations are a minimum. With rise in temperature, the saturation magnetization (M_s) diminishes gradually and then abruptly drops to zero at what is called the *Curie temperature* (T_c). Figure 14.32 shows the magnetization-temperature curves for Fe and Fe_3O_4 . The mutual spin coupling forces are completely destroyed at T_c , such that above T_c both ferro- and ferromagnetic substances are paramagnetic. The magnitude of Curie temperature varies from material to material, e.g. for Fe, Co, Ni and Fe_3O_4 the respective values of T_c are 768°C, 1120°C, 335°C and 585°C.

We may note that antiferromagnetism also affected by temperature. At Neel temperature (T_N), this behaviour vanishes and above T_N , antiferromagnetic materials also become paramagnetic.

11.6 Domains and Hysteresis

At a temperature below T_c , any ferro- or ferrimagnetic material is composed of small-volume regions in which there is a mutual alignment in the same direction of all magnetic moments (Fig. 14.33). Such a region is called domain, and each one domain is magnetized to its saturation magnetization. Adjacent domains are separated by *domain boundaries or walls*, across which the direction of magnetization gradually changes as shown in Fig. 14.34. As mentioned earlier, domains are normally microscopic in size, and for a polycrystalline specimen, each grain may consist of more than a single domain. Obviously, in a macro piece of specimen, there will be large number of domains, and all may have different magnetization orientations. Thus for the entire solid, the magnitude of the M field is the vector sum of the magnetizations of all the domains, each domain contribution being weighted by its volume fraction. We may note that for an unmagnetized specimen, the appropriately weighted vector sum of the magnetizations of all the domains is zero.

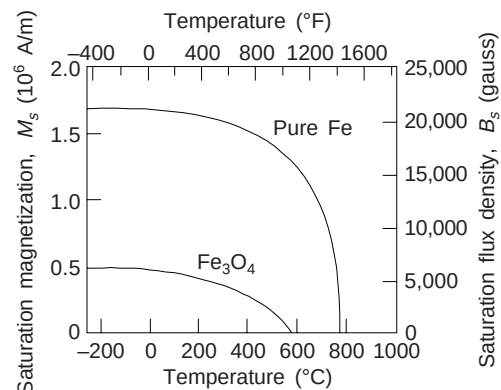


Fig. 14.32 Saturation magnetization (M_s) versus temperature curves for Fe and Fe_3O_4

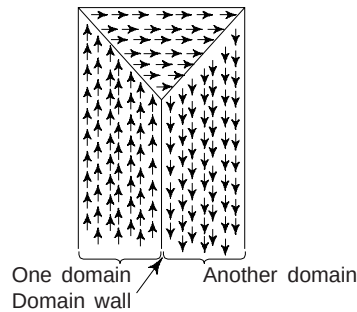


Fig. 14.33 Domains in a ferro- or ferrimagnetic material. The arrows represent atomic magnetic dipoles. Within each domain, all dipoles are aligned, whereas the direction of alignment varies from one domain to another domain

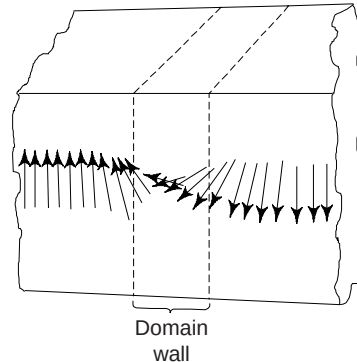


Fig. 14.34 The gradual change in magnetic dipole orientation across a domain wall or boundaries

When the ferromagnetic sample that is initially demagnetized is subjected to a continuously increasing magnetizing force H , the relation between H and flux density B is shown by the normal magnetization curve Oab (Fig. 14.35). The point a indicates the magnetic condition as the increasing magnetic intensity has reached H_1 . If H is increased to a maximum value H_2 and then decreased again to H_1 , the decreasing flux density does not follow the path of increase, but decrease at a rate less than that at which it rose. The lag in the change of B behind the change of H is called *hysteresis*. If the value of H is further reduced from H_1 to zero, B is not reduced to zero but to a value B_r . The specimen has retained a permanent magnetism. This ordinate B_r is called the *retentivity* or *remanence*. The value of B may be reduced to zero at e by reversing the direction of H and increasing its value of H_c . This value of H_c is called the *coercive force* or *coercivity*.

As H is increased in the negative direction, the magnetization proceeds along the curve of Fig. 14.35 until at f the values of B and H are the same as those at b , but opposite in direction. When reverse changes in H are made, the magnetization changes along the curve $fghb$. This entire loop $bdefghb$ is called a *hysteresis loop*. If the hysteresis loop starts from another point on the normal magnetization curve, such as a , there will be a smaller hysteresis loop entirely within the larger, such as the dotted loop of the Fig. 14.35. These curves sometimes called B - H curves and are used to describe magnetic materials. We may note that it is possible to reverse the direction of the magnetic field at any point along the curve and generate other hysteresis loops. To demagnetize the ferromagnetic or ferrimagnetic samples one has to repeatedly cycle it in H field that alternates direction and decreases in magnitude. Magnetization curves are generally obtained from long, thin samples to avoid demagnetization. The demagnetization effects are extremely important in ferromagnetic resonance.

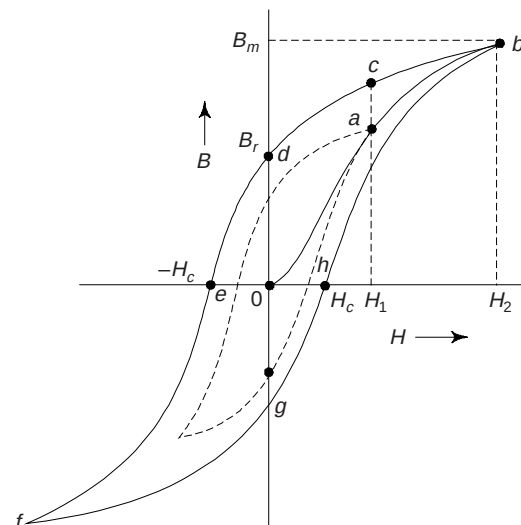


Fig. 14.35 Magnetic flux density (B) versus the magnetic field strength (H) for a ferromagnetic material that is subjected to forward and reverse saturations (points b and f). Hysteresis loop $b d e f g h b$. The remanence B_r and the coercive force H_c are also shown

We may note that ferromagnetic materials in common use are polycrystalline; i.e., a piece of the material consists of tremendous number of single crystals of random orientation. In each single crystal, there are many domains. The magnetization of the whole body consists of the magnetization of the various single crystals within the body.

In magnetizing the core, energy must be supplied. As H and B increase along $fg hb$ (Fig. 14.35), the energy gain is proportional to the area under that portion of the curve. Along the path $bcdef$, there is a loss in energy proportional to the area under $bcdef$. The net loss in energy per cycle per unit volume is given by Eq. (46) where

$$W = \oint dW = \oint H dB \quad (46)$$

the integral is taken around the closed loop. But $\oint H dB$ is the area of the hysteresis loop, and the energy loss per unit volume per cycle is equal to the area of the hysteresis loop. This energy is converted into heat. If B is expressed in teslas and H in Amp/m, the energy loss is in J/m^3 (cycle).

Steinmetz found an empirical relation between the energy loss per unit volume per cycle W and the maximum value B_m of the flux density during the cycle given by

$$W = \eta (B_m)^n \quad (47)$$

where η is called the *Steinmetz coefficient*. Steinmetz found a value of about 1.6 for the exponent n for many materials, but it varies from about 1.5 to 2.5 for others.

In alternating-current machinery, masses of iron are in fields that are constantly reversing. Therefore, the iron is constantly being carried around hysteresis paths, and there is an energy loss per cycle that depends upon the hysteresis loop for the particular iron that is used. This hysteresis loss results in undesirable heating of the iron as well as waste of energy. Obviously, the hysteresis curves are quite important in determining the quality of a magnetic material and selecting the material for a particular application. In accordance with their H_c and W values, ferromagnetic materials are classified as *soft* and *hard*.

11.6 Soft Magnetic Materials

Soft magnetic materials are characterized by their low loss and high permeability. There are a variety of alloys used with various combinations of magnetic properties, mechanical properties, and cost (Table 14.13). There are seven major groups of commercially important materials: iron and low carbon steels, iron-silicon alloys, iron-aluminium-silicon alloys, nickel-iron alloys, iron-cobalt alloys, ferrites and amorphous alloys.

Table 14.13 Some properties of selected soft magnetic materials

Material	Composition (wt%)	Initial relative permeability μ_i	Saturation flux density B_g [tesla (gauss)]	Hysteresis loss/cycle [J/m ³ (erg/cm ³)]	Resistivity ρ ($\Omega\cdot m$)
Commercial iron ingot	99.95Fe	150	2.14 (21,400)	270 (2700)	1.0×10^{-7}
Silicon-iron (oriented)	97Fe, 3Si	1400	2.01 (20,100)	40 (400)	4.7×10^{-7}
45 Permalloy	55Fe, 45Ni	2500	1.60 (16,000)	120 (1200)	4.5×10^{-7}
Supermalloy	79Ni, 15Fe, 5Mo, 0.5Mn	75,000	0.80 (8000)	—	6.0×10^{-7}
Ferroxcube A	48MnFe ₂ O ₄ 52ZnFe ₂ O ₄	1400	0.33 (3300)	~40 (~400)	2000
Ferroxcube B	36NiFe ₂ O ₄ 64ZnFe ₂ O ₄	650	0.36 (3600)	~35 (~350)	10^7

The behaviour of soft materials is controlled by the pinning of domain walls at heterogeneities such as grain boundaries and inclusions. In addition, eddy-current loss is minimized through alloying additions which increase the electrical resistivity. Initial permeability, important in electronic transformers and inductors, is improved by minimizing all sources of magnetic anisotropy. A high maximum permeability, necessary for motors and power transformers, is increased by the alignment of the anisotropy, for e.g., through development of crystal texture or magnetically induced anisotropy. For this reason the relative area within the hysteresis loop must be small; it is characteristically thin and narrow (Fig. 14.36).

The saturation field or magnetization is determined only by the composition of the material. For e.g., in cubic ferrites, substitution of a divalent metal ion such as Ni^{2+} for Fe^{2+} in $\text{FeO} - \text{Fe}_2\text{O}_3$ will change the saturation magnetization. However, susceptibility and coercivity (H_c), which also influence the shape of the hysteresis curve, are sensitive to structural variables rather than to composition. We may note that a low value of coercivity corresponds to the easy movement of domain walls as the magnetic field changes magnitude and/or direction. Structural defects, e.g. particles of a non magnetic phase or voids in the magnetic material tend to restrict the motion of domain walls, and thus increase the coercivity (H_c). Obviously, a soft magnetic material must be free from such structural defects.

Electrical resistivity is the another property which need to be considered. In addition to hysteresis energy losses, energy losses may result from electrical currents that are induced in a magnetic material by a magnetic field that varies in magnitude and direction with time; these are called *eddy currents*. These energy losses in soft magnetic materials can be minimized by increasing the electrical resistivity. This is achieved in ferromagnetic materials by forming solid solution alloys; Fe-Si and Fe-Ni alloys, etc. The ceramic ferrites are commonly used for applications requiring soft magnetic materials because they are intrinsically insulators. However, their applicability is somewhat limited, in as much as they have relatively small susceptibilities.

The class of alloys used in largest volume is by far Fe and 1-3.5% Si-Fe for applications in motors and large transformers. In these applications the cost of the material is often the dominant factor, with losses and excitation power secondary but still important. A major improvement in these alloys occurred around 1940 with a development of a $\{110\} \langle 001 \rangle$ crystal texture in 3.2% Si-Fe which greatly reduced losses and increased permeability. Since then the losses have decreased steadily, roughly logarithmically, through alloying, texture improvements, decreasing the thickness of strip and application of stressed insulating coatings.

Many special alloys find their use in special devices designed to exploit unusual properties. These include materials with high saturation induction B_s (the magnetic induction at very large values of H), usually Co-Fe alloys; and alloys with high μ , most often Ni-Fe alloys. These alloys are the mainstay of the telecommunication industry. Some of the devices require high initial permeability (μ at very low fields) and other may depend on high values of μ_{max} (the maximum permeability).

In power equipment the major consideration is the power lost in the magnetic circuit under operating conditions. The total core loss P_c consists of the hysteresis loss P_h and eddy current loss P_e . Thus

$$P_c = P_h + P_e = \text{const} \times f + \text{const} \times \frac{B^2 f^2}{\rho} \quad (48)$$

where f is the frequency and ρ is the electrical resistivity of the material.

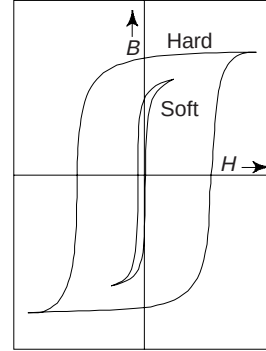


Fig. 14.36 Magnetization curves for soft and hard magnetic materials

The hysteresis loss may be kept low by using a material with a narrow hysteresis loop (one with low coercive force H_c , the value of H required to reduce B to zero) and minimizing mechanical strain after the final stress-relief anneal. Eddy-current loss may be reduced by breaking up eddy-current paths, for e.g. by using laminations rather than solid cores.

Audio-frequency devices require very thin laminations or resin-bonded alloy powder cores. The very high electrical resistivity ($1 - 10^6 \Omega\text{-cm}$) qualifies many ferrites for very high frequency applications, for e.g., in cores and transformers.

It is interesting to note that the hysteresis characteristics of soft magnetic materials may be enhanced for some applications by an appropriate heat treatment in the presence of a magnetic field. Using such a technique, one may produce a square hysteresis loop, which is desirable in some magnetic amplifier and pulse transformer applications. Moreover soft magnetic materials are used in switching circuits.

11.7 Hard Magnetic Materials

Hard magnetic materials are utilized in permanent magnets, which have a high resistance so that they strongly resist demagnetization once magnetized. In view of hysteresis behaviour, a *hard magnetic material* or permanent magnet has a high remanence, coercivity, and saturation flux density, as well as a low initial permeability, and high hysteresis energy losses. Figure 14.36 show a comparison of the hysteresis characteristics of hard and soft magnetic materials.

The important characteristics relative to applications for hard magnetic materials are the coercivities H_c and what is termed the “energy product” $(BH)_{\max}$. The energy product $(BH)_{\max}$ corresponds to the area of the largest B - H rectangle that one can construct within the second quadrant of the hysteresis curve (Fig. 14.37). The units of $(BH)_{\max}$ are kJ/m^3 (MGOe: $1 \text{ MGOe} = 10^6 \text{ gauss-oersted} = 7.96 \text{ kJ/m}^3$). We may note that the value of $(BH)_{\max}$ is representative of the energy required to demagnetize a permanent magnet, i.e., the larger $(BH)_{\max}$ the harder is the magnetic material in terms of its magnetic characteristics. Although the overall quality of a permanent magnet depends on $(BH)_{\max}$, but the design considerations, high H_c , high residual induction B_r (the magnetic induction when H is reduced to zero) and reversibility of permeability μ may also be controlling factors.

We may note that hysteresis behaviour of a material is related to the ease with which the magnetic domain boundaries move; by impeding domain wall motion, the coercivity and susceptibility are enhanced, such that a large external field is required for demagnetization. Moreover, these characteristics are interrelated to the microstructure of the material. To understand the relation between the resistance to demagnetization, i.e., the coercivity H_c and the metallurgical microstructure, it is essential to understand the mechanisms of magnetization reversal. The two major mechanisms are reversal against a *shape anisotropy* and reversal through nucleation and growth of reverse magnetic domains against *crystal anisotropy*.

Conventional Hard Magnetic Materials

Hard magnetic materials can be studied under two main categories: *Conventional* and *high energy hard materials*. The conventional hard magnetic materials have $(BH)_{\max}$ values that range between about 2 and 80 kJ/m^3 (0.25 and 10 MGOe). Ferromagnetic materials—magnet steels, cunife (Cu-Ni-Fe) alloys, alnico (Al-Ni-Co) alloys as well as hexagonal ferrites ($\text{BaO-6Fe}_2\text{O}_3$) are conventional hard mag-

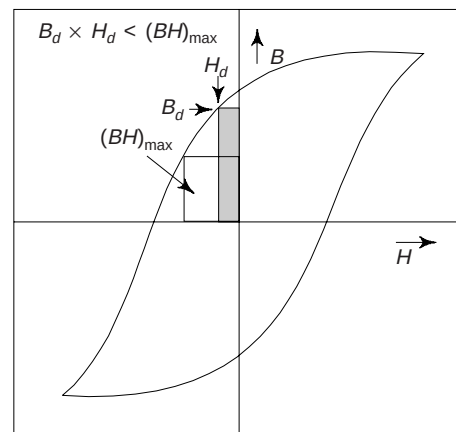


Fig. 14.37 Magnetization curve displaying hysteresis. Within the second quadrant, two B - H energy product rectangles are drawn, the area of rectangle labelled $(BH)_{\max}$ is the largest possible, which is greater than the area defined by $B_d - H_d$

netic materials. Some of the critical properties of few of these hard magnetic materials are presented in Table 14.14.

Table 14.14 Representative properties of few hard magnetic materials

Material	Composition (wt%)	Remanence B_r [tesla (gauss)]	Coercivity H_c [amp-turn/m (Oe)]	$(BH)_{max}$ [kJ/m ³ (MGOe)]	Curie temperature T_c [°C (°F)]	Resistivity ρ (Ω -m)
Tungsten steel	92.8 Fe, 6 W, 0.5 Cr, 0.7 C	0.95 (9500)	5900 (74)	2.6 (0.33)	760 (1400)	3.0×10^{-7}
Cunife	20 Fe, 20 Ni, 60 Cu	0.54 (5400)	44,000 (550)	12 (1.5)	410 (770)	1.8×10^{-7}
Sintered alnico 8	34 Fe, 7 Al, 15 Ni, 35 Co, 4 Cu, 5 Ti	0.76 (7600)	125,000 (1550)	36 (4.5)	860 (1580)	—
Sintered ferrite 3	BaO-6Fe ₂ O ₃	0.32 (3200)	240,000 (3000)	20 (2.5)	450 (840)	$\sim 10^4$
Cobalt rare earth 1	SmCo ₅	0.92 (9200)	720,000 (9,000)	170 (21)	725 (1340)	5.0×10^{-7}
Sintered neodymium iron-boron	Nd ₂ Fe ₁₄ B	1.16 (11,600)	848,000 (10,600)	255 (32)	310 (590)	1.6×10^{-6}

The hard magnet steels are generally alloyed with tungsten and/or chromium. These two elements readily combine with carbon in the steel to form tungsten and chromium carbide precipitate particles, which are especially effective in obstructing domain wall motion under the proper heat-treating conditions. An appropriate heat treatment for the other metal alloys forms extremely small single domain and strongly Fe-Co particles within a non magnetic matrix phase.

High-Energy Hard Magnetic Materials

Recently, some intermetallic compounds having a variety of compositions have been developed. These permanent magnetic materials have $(BH)_{max}$ in excess of about 80 kJ/m³. The two that have found commercial exploitation are SmCo₅ and Nd₂Fe₁₄B (Table 14.14).

Samarium-Cobalt Permanent Magnets (SmCo₅)

SmCo₅ is a member of a group of alloys that are combinations of Co or Fe and a light rare-earth element; a number of these alloys exhibit high-energy, hard magnetic behaviour, but SmCo₅ is the only one material that is of commercial significance. $(BH)_{max}$ for SmCo₅ materials [between 120 and 240 kJ/m³ (15 and 30 MGOe)] are quite high in comparison to conventional hard magnetic materials (Table 14.14); in addition, they have relatively large H_c (coercivities). One uses powder metallurgical techniques to fabricate SmCo₅ magnets. First, the appropriately alloyed material is ground into a fine powder and then the powder particles are aligned using an external magnetic field and thereafter pressed into the desired shape. The piece is then sintered at an elevated temperatures, followed by another heat treatment which improves the magnetic properties of the material.

Neodymium-Iron-Boron Permanent Magnets (Nd₂Fe₁₄B)

Sm is a rare and relatively expensive material. Moreover, the price of Co is variable and its sources are unreliable. This is why, the Nd₂Fe₁₄B alloys have become the materials of choice for a large number and wide diversity of applications requiring hard magnetic materials. We may note that H_c and $(BH)_{max}$ of these materials rival those of SmCo₅ alloys (Table 14.14).

The magnetization and demagnetization behaviour of these hard magnetic materials is a function of domain wall mobility, which, in turn, is controlled by the final microstructure-i.e., the size, shape and the orientation of the crystallites and grains, as well as the nature and distribution of any second-phase particles that are present. We may note that microstructure will depend on how the material is processed. There are two different processing techniques available for the fabrication of $\text{Nd}_2\text{Fe}_{14}\text{B}$ magnets: *powder metallurgy* (sintering) and *rapid solidification* (melt spinning). The powder metallurgical method is similar to that described for the SmCo_5 materials. In case of rapid solidification techniques, the alloy, in molten form, is quenched very rapidly such that either an amorphous or very fine grained and thin ribbon is produced. This ribbon material then pulverized, compacted into the shape as per requirement, and subsequently heat treated. Rapid solidification is the more involved of the two fabrication processes; nevertheless, it is continuous, whereas powder metallurgy is batch process, which has its inherent disadvantages.

Hard materials are used in motors, loudspeakers, meters, holding devices and in a host of different devices in a variety of technological fields. Permanent magnets are far superior to electromagnets in that their magnetic fields are continuously maintained and without the necessity of having to expend electrical power. Moreover, no heat is generated during operation. This is why permanent magnets are preferred in motors. Motors with permanent magnets are much smaller than their electromagnet counterparts and are utilized extensively in fractional horsepower units. Few familiar motor applications include: in cordless drills and screw drivers; in automobiles (starting, window wiper, washer and fan motors); in clocks and in audio and video recorders. There are other common devices, e.g. speakers in audio systems, light-weight earphones, hearing aids, and computer peripherals etc. that employ these hard magnetic materials.

11.8 Magnetic Storage

Recently, magnetic materials have become increasingly important in the field of *information storage*; in fact, magnetic recording has become virtually the universal technology for the storage of electronic information. The audio tapes, VCRs, computer hard disks, floppy disks, credit cards, etc. are few examples of magnetic storage. In computers, semiconductor elements serve as primary memory and magnetic disks are used for secondary memory because they are capable of larger quantities of information and also at a lower cost. The recording and television industries rely heavily on magnetic tapes for the storage and reproduction of audio and video sequences.

Computer bytes, sound, or visual images in the form of electrical signals are recorded on very small segments of the magnetic storage device-a tape or disk. Transference to and retrieval from the magnetic tape or disk is accomplished by means of an inductive read-write head, which consists basically of a wire coil wound around a magnetic material core into which a gap is cut. Using electrical signal, data are introduced (or “written”) within the coil, which generates a magnetic field across the gap. This magnetic field in turn magnetizes a very small area of the disk or tape within the proximity of the head. Upon removal of the magnetic field, the magnetization remains; i.e. the signal has been stored. Figure 14.38 shows the essential features of this recording process.

We may note that the same head may be utilized to retrieve (or “read”) the stored information. A voltage is induced when there is a change in the

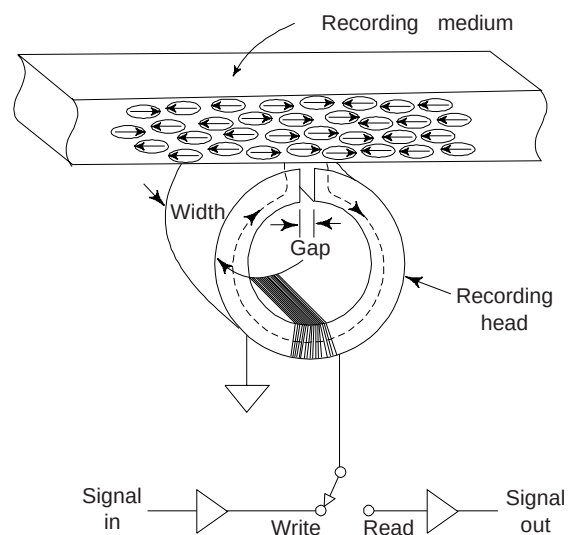


Fig. 14.38 Using magnetic storage medium, how information is stored and retrieved

magnetic field as the tape or disk passes by the head coil gap. One can amplify this and then converted back into its original form or character (Fig. 14.38).

More recently, hybrid heads that consist of an inductive-write and a magnetoresistive read head in a single unit have been introduced. In the magnetoresistive head, the electrical resistance of the magnetoresistive thin film element is changed as a result of magnetic field changes when the tape or disk passes by the read head. We may note that higher sensitivities and higher data transfer rates make magnetoresistive very attractive.

Magnetic media are of two principal types: *particulate* and *thin film*. First media consist of very small needle-like or acicular particles, normally of γ -Fe₂O₃ ferrite or CrO₂; these are applied and bonded to a polymeric film (for magnetic tapes) or to a metal or polymer disk. During manufacture, these particles are aligned with their long axis in a direction that parallels the direction of motion past the head. Each particle is a single domain that may be magnetized only with its magnetic moment lying along this axis. Corresponding to the saturation magnetization in one axial direction, and its opposite, two magnetic states are possible. These two states make possible the storage of information in digital form, as 1's and 0's. In one system, a 1 is represented by a reversal in the magnetic field direction from one small area of storage medium to another as the numerous acicular particles of each such region pass by the head. A lack of reversal between adjacent region is indicated by a 0.

The *thin-film storage technology* provides higher storage capacities at lower costs. It is used mainly on rigid disk drives and consists of multilayered structure. A magnetic thin film layer is the actual storage component. This thin film is normally either a CoPtCr or CoCrTa alloy, with a thickness of between 10 and 50 nm (100 and 500 Å). A substrate layer below and upon which the thin film resides is pure Cr or Cr alloy. The thin film is itself is polycrystalline, having an average grain that is typically between 10 and 30 nm (100 and 300 Å). Within the thin film each grain is a single magnetic domain and it is highly desirable that grain shape and size be relatively uniform. For magnetic storage disk where these thin films are employed, the crystallographic direction of easy magnetization for each grain is aligned in the direction of disk motion (or the direction opposite). We may note that the mechanism of magnetic storage within each of these single domain grains is the same for the needle shaped particles, i.e. the two magnetic states correspond to domain magnetization in one direction or its antiparallel equivalent.

We may note that the storage density of thin films is greater than for particulate media because the packing efficiency of thin-film domains is greater than for the acicular particles; particles will always be separated with void space in between. The known storage densities for particulate media are of the order of 1×10^8 bit/in² (1.5×10^5 bit/mm²). For thin films, storage densities are approximately an order of magnitude greater, i.e. $\sim 5 \times 10^{10}$ bit/in² (8×10^7 bit/mm²).

The hysteresis loops for these magnetic storage media should be relatively large and square. Obviously, these characteristic ensure that storage will be permanent, and in addition, magnetization reversal will result over a narrow range of applied field strengths. For particular recording media, saturation flux density normally ranges from 0.4 to 0.6 tesla. For thin films, B_s lie between 0.6 and 1-2 tesla. H_c values are typically in the range of 1.5×10^5 to 2.5×10^5 A/m, i.e. 2000 to 3000 °C.

Example 6 The saturation magnetization of BCC Iron is 1750 kA/m. Show that the net magnetic moment per iron atom in the crystal is 2.2. Given lattice parameter of BCC iron = 2.87 Å [B.E., M.Sc.(M.S.)]

Solution Volume of unit cell of BCC iron = $(2.87)^3 \times 10^{-30}$ m³

Number of atoms in the unit cell = 2

$$\therefore \text{Net magnetic moment per atom} = 1750 \times 1000 \times (2.87)^3 \times 10^{-30} \times 1/2 \\ = 2.068 \times 10^{-23} \text{ A-m}^2$$

$$\text{The magnetic moment (in units of } \mu_B) = \frac{2.068 \times 10^{-23}}{9.273 \times 10^{-24}} = 2.2$$

Example 7 The density of nickel is $8.90 \times 10^3 \text{ kg/m}^3$. Avogadro's number $N_A = 6.023 \times 10^{23} \text{ atoms/mol}$. Atomic weight of Ni is 58.71 gm/mol . Calculate (i) the saturation magnetization (ii) the saturation flux density. [B.E., Diploma]

Solution

(i) $M_s = 0.60 \mu_B N$

$$\begin{aligned} \text{and } N (\text{number of atoms/m}^3) &= \frac{\rho N_A}{A} \\ &= \frac{8.90 \times 10^6 \text{ gm/m}^3 \times 6.023 \times 10^{23} \text{ atoms/mol}}{58.71 \text{ gm/mol}} \\ &= 9.13 \times 10^{28} \text{ atoms/m}^3 \\ \therefore M_s &= \left(\frac{0.60 \text{ Bohr magneton}}{\text{atom}} \right) \left(\frac{9.27 \times 10^{-24} \text{ A-m}^2}{\text{Bohr magneton}} \right) \left(\frac{9.13 \times 10^{28} \text{ atoms}}{\text{m}^3} \right) \\ &= 5.1 \times 10^5 \text{ A/m} \end{aligned}$$

(ii) Saturation flux density,

$$\begin{aligned} B_s &= \mu_o M_s \\ &= \left(\frac{4\pi \times 10^{-7} \text{ H}}{\text{m}} \right) \left(5.1 \times 10^5 \frac{\text{A}}{\text{m}} \right) \\ &= 0.64 \text{ tesla.} \end{aligned}$$

Example 8 Each cubic unit cell of Fe_3O_4 contains 8 Fe^{2+} and 16 Fe^{3+} ions and the unit cell edge length is 0.839 nm . Calculate the saturation magnetization of Fe_3O_4 . [B.E., M.Sc. (M.S.)]

Solution

$$M_s = N' \mu_B \quad (\text{i})$$

where N' is the number of Bohr magneton per unit cell n_B divided by the unit cell volume, V_C , i.e.

$$N' = \frac{n_B}{V_C} \quad (\text{ii})$$

We may note that the net magnetization results from Fe^{2+} ions only. Since there are 8 Fe^{2+} ions per unit cell and 4 Bohr magnetons per Fe^{2+} ion, therefore, $n_B = 32$. Since unit cell is a cube, and therefore $V_C = a^3$, a being the unit cell edge length. Therefore,

$$\begin{aligned} M_s &= \frac{n_B \mu_B}{a^3} \\ &= \frac{(32 \text{ Bohr magneton})}{\text{unit cell}} \frac{9.27 \times 10^{-24} \text{ A-m}^2}{\text{Bohr magneton}} \\ &= \frac{(0.839 \times 10^{-3} \text{ m})^3 / \text{unit cell}}{} \\ &= 5 \times 10^5 \text{ A/m.} \end{aligned}$$

Example 9 In a 440 V, 50 Hz transformer, the total iron loss is 2300 W. When the applied voltage is 220 V at 25 Hz, the total iron loss is 750 W. Show that the eddy current loss at the normal voltage and frequency is 1600 W. [BE]

Solution We know that the hysteresis loss (P_h) and the induced *emf* loss (P_e) are proportional to the frequency. P_e is proportional to the square of the induced *emf*. We have

$$P_e + P_h = 750 \text{ W (at 25 Hz)}$$

At 440 V and 50 Hz, we have

$$4P_e + 2P_h = 2300 \text{ W (at 50 Hz)}$$

or

$$2P_e = 800 \text{ W}$$

∴ The eddy current loss at the normal voltage and frequency = $4P_e = 1600 \text{ W}$.

SUGGESTED READINGS

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REVIEW QUESTIONS

1. Distinguish between a metal, a semiconductor and an insulator on the basis of their energy band structure.
2. (a) Explain electrical conductivity on the basis of energy band diagram.
(b) What is the importance of a energy band diagram and how does it compare the conductor and insulator? [AMIE]
3. What is an insulator? What is the reason that a material behaves an insulator? [AMIE]
4. Describe an expression for the conductivity of conductors from the first principle. [AMIE]
5. Discuss factors that affect the electrical resistance of materials. [AMIE]
6. Explain what insulators, conductors and semiconductors are? Give examples of each. [AMIE]
7. Define a dielectric. Explain the common properties and uses of dielectrics. [AMIE]
8. Describe with examples the three categories of dielectrics. [AMIE, Diploma]
9. Describe the characteristic properties of ferroelectric materials.
10. Explain the meaning and origin of piezoelectricity. Justify the statement that “all ferroelectric crystals are piezoelectric, but all piezoelectric crystals are not necessarily ferroelectric”.
11. How do temperature and impurities affect electrical resistivity of metals? [AMIE]
12. Distinguish between the characteristics of dia, para, ferro, ferri and antiferro magnetism. Give an example of each type of material. Comment on the temperature variation of susceptibility of all types of materials.
13. Classify the magnetic materials. [AMIE]
14. Describe briefly paramagnetism, diamagnetism and ferromagnetism. [Diploma]
15. Explain: (i) Diamagnetism (ii) paramagnetism (iii) ferromagnetism (iv) antiferromagnetism and (v) ferrimagnetic materials. [AMIE]
16. Compare properties of diamagnetic, paramagnetic and ferroelectric materials. [AMIE]
17. Draw B - H curve for ferromagnetic materials and explain it. [MSU, AMIE]
18. Explain the principle of a ferrimagnetism with a diagram. [AMIE]
19. Describe the effects of magnetostriction on the properties of some materials. [BE]
20. How can a soft material be converted into a hard one? Distinguish between the two types, giving examples. What are their main applications? [BHU (Elect.), AMIE]
21. Write short note on the following
 - (i) Ferroelectricity
 - (ii) Piezoelectricity
 - (iii) Electrostriction
 - (iv) Dielectrics

- (v) Ferrites (vi) Soft and hard magnets
 (vii) B-H curve for ferromagnetic materials (viii) Hard magnetic materials
 (ix) Magnetization (x) Magnetic domains
 (xi) Curie point (xii) Ferromagnetism
 (xiii) Uses of dielectric materials (xiv) Soft ferrites
 (xv) Antiferromagnetism (xvi) Ferrimagnetism [AMIE]
22. Write whether *True* or *False* and justify your answer:
 (a) Presence of phosphorus increases conductivity of copper.
 (b) Increase in temperature of a conductor increases resistivity.
 (c) Elevated service temperature decreases the dielectric strength of insulating materials.
 (d) Conductivity of metals decreases at elevated temperature.
 (e) For permanent magnetic materials, the product (BH) should be maximum possible. [MSU, Diploma]
23. Explain, why:
 (a) In actual practice, several thin layers of dielectric are used in capacitors instead of a single thick layer?
 (b) Oxygen free high conductivity copper is specified for bus bars?
 (c) Manufacturers usually specify maximum safe operating temperature on electric motors?
 (d) In case of dielectric materials, energy required for an electron to cross the gap is large?
 (e) The breakdown voltage does not increase in proportion to the increase in thickness of dielectrics? [Diploma, M.Sc. (M.S.)]
24. Which of the two elements, cobalt and gadolinium, has the higher saturation magnetization at (i) 0 K, and (ii) 300 K? [B.E.]
- [Hint:**
 (i) The thermal energy at 0 K is kT . This tends to randomize the spins and hence thermal energy is zero. Obviously, all the spins remain aligned in both Co and Gd. The net magnetic moment of Gd is 7/atom and that of Co is only 1.7 per atom. This means Gd will have higher saturation magnetization. In fact, the actual values will depend on the number of atoms per unit volume in the two cases.
 (ii) Curie temperature of Gd is 289 K. Obviously, Gd at 300 K will be paramagnetic and will have negligible magnetization as compared to Co, which has a much higher curie temperature.]
25. In NiFe_2O_4 , the ferric ions are antiferromagnetically coupled. The magnetization is due to the Ni ions. When Zn is added to NiFe_2O_4 , the magnetization increases, even though the zinc ions are not ferromagnetic. Explain? [M.Sc. (M.S.), Diploma]
- [Hint:** Looking to the crystal structure of NiFe_2O_4 crystal, we note that it has inverse spinel structure and all the Ni^{2+} and half of the Fe^{3+} ions in the octahedral sites. The remaining other half of the Fe^{3+} are in tetrahedral sites. The ferric ions in the two sites are antiferromagnetically coupled. When Zn is substituted for the ferric ions in the tetrahedral sites, the antiferromagnetic coupling will not exactly cancel because the number of ferric ions in the octahedral sites will be more. Obviously, this would increase the magnetization, as there is an extra contribution from the uncoupled ferric ions of the octahedral sites.]
26. What is the distinction between electronic and ionic conduction? [B.E.]
27. How does the electron structure of an isolated atom differ from that of a solid material? [AMIE]
28. If a material is cooled through its melting temperature at an extremely rapid rate, it will form a non-crystalline solid, i.e., a *metallic glass* (Metallic glasses are formed from reasonably inexpensive metals like Fe, Ni, Co and have a number of superior properties for technical applications compared to those possessed by the crystalline solids, for e.g., the filaments of metal glasses are stronger and stiffer and hence offer potential application as lost cost high performance structural reinforcement elements).

29. (a) Explain the two sources of magnetic moments for electrons.
 (b) Do all electrons have a net magnetic moment? Why or why not?
 (c) Do all atoms have a net magnetic moment? Why or why not? [B.E., Diploma]
30. There is associated with each atom in paramagnetic and ferromagnetic materials a net magnetic moment. Explain why ferromagnetic materials can be permanently magnetized whereas paramagnetic ones cannot. [AMIE]
31. Mention the major similarities and differences between ferromagnetic and ferrimagnetic materials. [BHU(BE)]
32. What is the difference between the *spinel* and *inverse spinel* crystal structures? [B.E.]
 [Hint: The cubic ferrites have the *spinel* crystal structure. There are eight occupied tetrahedral (A) sites and 16 occupied octahedral (B) sites in a unit cube. The lattice constant is about 8 Å. A remarkable feature of the spinels is that all exchange integrals J_{AA} , J_{AB} and J_{BB} are negative and favour *antiparallel alignment* of the spins connected by the interaction.]
33. Mention the differences between hard and soft magnetic materials in terms of both hysteresis behaviour and typical applications. [B.E., Diploma]
34. It is possible, by various means, e.g., alteration of microstructure and impurity additions, to control the ease with which domain walls move as the magnetic field is changed for ferro and ferri magnetic materials. Draw and *B-H* hysteresis loop for a ferromagnetic material, and superimpose on this plot the loop alterations that would occur if domain boundary movement were hindered. [M.Sc. (M.S.), B.E.]

PROBLEMS

- The resistance of a given sample of copper of diameter 0.1 mm and length 0.2 m is 0.439Ω . Calculate its resistivity and conductivity. [Ans. $1.72 \times 10^{-8} \Omega - \text{m}$, $5.81 \times 10^7 / \Omega - \text{m}$] [B.E.]
- At room temperature the electrical conductivity and the electron mobility for Cu are $6 \times 10^7 / (\Omega - \text{m})$ and $0.0030 \text{ m}^2 / (\text{V} - \text{s})$, respectively. (a) Find the number of free electrons per m^3 for Cu at room temperature. (b) What is the number of free electrons per Cu atom? Given density 8.9 gm/cm^3 .

$$\left[(a) n = 1.25 \times 10^{29} / \text{m}^3, (b) 1.48 \frac{\text{free electrons}}{\text{atom}} \right] \quad [\text{AMIE, B.E.}]$$
- Show that the percentage increase in resistivity of (i) Cu (ii) nichrome on heating from 300 K to 1000 K will be 300% and 7% respectively. [AMIE]
- Determine the power loss due to hysteresis in transformer core of 0.01 m^3 volume of 50 Hz frequency. The area of the loop is 600 J/m^3 . [AMIE]
- The unit cell of magnetite is cubic with a lattice parameter of 8.37 Å . There are 16 ferric ions and 8 ferrous ions in the unit cell. Show that the saturation magnetization of magnetite is 510 kA/m . [B.E.]
- The electrical resistivity of iron increases by a factor of 6, on alloying with 4% Si. Show that the eddy current losses for this change in composition is 83%. [AMIE]
- A coil of wire 0.20 m long and hanging 200 turns carries a current of 10 Å . (a) Find the magnitude of the magnetic field strength H (b) If the coil is in vacuum then compute the flux density B (c) If the susceptibility for titanium is 1.81×10^{-4} (SI units), then calculate the flux density inside a bar of titanium that is positioned within the coil (d) Calculate the magnetization M . [B.E.]
 (a) $H = 10,000 \text{ A-turns/m}$, (b) $B_0 = 1.257 \times 10^{-2} \text{ tesla}$,
 (c) $B \cong 1.257 \times 10^{-2} \text{ tesla}$ (d) $M = 1.81 \text{ A/m}$
- The chemical formula for manganese ferrite can be expressed as $(\text{MnFe}_2\text{O}_4)_8$ because there is eight formula units per unit cell. If this material has a saturation magnetization of $5.6 \times 10^5 \text{ A/m}$ and density 5 gm/cm^3 , find the number of Bohr magnetons associated with each Mn^{2+} ion.
 $[4.6 \text{ Bohr Magnetons/Mn}^{2+} \text{ ion}] \quad [\text{B.E., M.Sc. (MS)}]$

9. On alloying with 4% silicon, the electrical resistivity of iron increases by a factor of 6. Show that the decrease in eddy current losses for this change in composition is 83%. [B.E.]

OBJECTIVE QUESTIONS

1. Most widely used conducting materials are
 - (a) gold and silver
 - (b) tungsten and platinum
 - (c) copper and aluminium
 - (d) silicon and germanium
2. The SI units of electrical conductivity are
 - (a) $\Omega\text{-m}$
 - (b) $\Omega^{-1}\text{-m}^{-1}$
 - (c) $\text{kg}^{-1}\text{ m}^{-3}\text{ s}^3\text{A}^2$
 - (d) Ω/m
3. Ohm's law relates the electric field E , conductivity σ and current density J as
 - (a) $J = \sigma E$
 - (b) $J = \sigma E^{1/2}$
 - (c) $J = \sigma E^2$
 - (d) $J = E/\sigma$
4. If e , μ and n respectively represent the charge, mobility and concentration of electrons respectively, then the electrical conductivity of a metal is
 - (a) $ne\mu$
 - (b) $ne^2\mu$
 - (c) $n^2e^2\mu$
 - (d) ne/μ
5. The Fermi level for copper is 7 eV. The maximum velocity of free electrons at 0 K is
 - (a) 0 km/s
 - (b) 1110 km/s
 - (c) 1570 km/s
 - (d) 7000 km/s
6. If ρ_o and $\rho(T)$ are the temperature independent and temperature dependent parts of the resistivity of a metal, then the total resistivity is
 - (a) $\rho = \rho_o - \rho(T)$
 - (b) $\rho = \rho_o + \rho(T)$
 - (c) $\rho = \rho_o + (\rho(T))^2$
 - (d) $\rho = \rho_o/\rho(T)$
7. Which of the following metals has the lowest temperature coefficient of resistance?
 - (a) Silver
 - (b) Aluminium
 - (c) Gold
 - (d) Copper
8. An insulator is a material whose resistivity is approximately within the range
 - (a) $10^7\ \Omega\text{-m}$ to $10^{13}\ \Omega\text{-m}$
 - (b) $10^{-3}\ \Omega\text{-m}$ to $10^6\ \Omega\text{-m}$
 - (c) $10^3\ \Omega\text{-m}$ to $10^{17}\ \Omega\text{-m}$
 - (d) $10^{-9}\ \Omega\text{-m}$ to $10^9\ \Omega\text{-m}$
9. The Fermi level E_F depends on the length L of a linear solid as
 - (a) E_F is independent of L
 - (b) $L^{1/2}$
 - (c) L^2
 - (d) L^3
10. Volt in SI units is
 - (a) $\text{kgm}^2\text{s}^{-3}\text{A}^{-1}$
 - (b) $\Omega\text{-A}$
 - (c) $\Omega\text{-A}^{-1}$
 - (d) $\text{Nm}^2\text{s}^{-1}\text{A}^{-1}$
11. If the Fermi energy of a metal is 1.4 eV, the Fermi temperature of metal is approximately
 - (a) $1.6 \times 10^2\ \text{K}$
 - (b) 16 K
 - (c) $1.6 \times 10^3\ \text{K}$
 - (d) $1.6 \times 10^4\ \text{K}$
12. The composition of manganin alloy is
 - (a) 60% Cu, 10% Mn, 30% Ni
 - (b) 80% Cu, 10% Mn, 10% Ni
 - (c) 80% Cu, 20% Mn
 - (d) 60% Cu, 15% Mn, 25% Ni
13. In order to increase the mechanical strength of an aluminium conductor, one should go for
 - (a) doping
 - (b) cold working
 - (c) heat treatment
 - (d) steel reinforcement
14. Magnetic susceptibility has the dimensions of
 - (a) Wb-m
 - (b) dimensionless
 - (c) Wb/m^2
 - (d) Amp/m
15. 1 Bohr magneton is
 - (a) $9.27 \times 10^{-13}\ \text{amp-m}^2$
 - (b) $9.27 \times 10^{24}\ \text{amp-m}^2$
 - (c) $9.27 \times 10^{-24}\ \text{amp-m}^2$
 - (d) $9.27 \times 10^{-6}\ \text{amp-m}^2$
16. The units of magnetic permeability are
 - (a) $\text{Wb-A}^{-1}\text{-m}^{-1}$ (or H/m)
 - (b) Wb/m^2
 - (c) A/m
 - (d) None of these
17. In SI units, tesla is expressed as
 - (a) $\text{kg/m}^2\text{s}^2\text{A}$
 - (b) $\text{kg/s}^2\text{A}$
 - (c) $\text{NA}^{-1}\text{ m}^{-3}$
 - (d) none of the above

18. Magnetic induction B and magnetic field intensity H are related by
 (a) $B = \mu_o + \mu_o H$ (b) $B = \mu_o \mu_r H$
 (c) $B = \mu_o H^2$ (d) $B = \mu_o + H$ (b)
19. The magnetization (M) of a solid, magnetic induction B and field strength H are related by
 (a) $M = (B/\mu_o) - H$ (b) $M = 1 + B/H$
 (c) $M = 1 - \frac{B}{H}$ (d) $M = B - \mu_o H$ (a)
20. Which one of the following material does not have permanent magnetic dipoles?
 (a) ferromagnetic (b) antiferromagnetic
 (c) paramagnetic (d) diamagnetic (d)
21. A suitable material for audio and TV-transformers is
 (a) pure iron (b) Fe and 4% Si alloy
 (c) Iron and 30% Ni alloy (d) Ferrite (d)
22. The highest measured energy product in a permanent magnet is about
 (a) 200 kJ/m³ (b) 300 kJ/m³ (c) 400 kJ/m³ (d) 600 kJ/m³ (c)
23. In a ferromagnetic material, susceptibility is
 (a) very large and negative (b) very large and positive
 (c) zero (d) negative (b)
24. For Cu the resultant spin in Bohr magneton is
 (a) 5 (b) 9 (c) 3 (d) 0 (d)
25. Below the ferromagnetic curie temperature, the ferromagnetic material exhibit $B-H$ whose form is
 (a) straight line (b) exponential curve
 (c) $B-H$ loop (d) none of these (c)
26. The temperature below which certain materials are ferromagnetic and above which they are paramagnetic is called
 (a) Neel temperature (b) Curie temperature
 (c) Weiss temperature (d) None of these (b)
27. At curie temperature, the spontaneous magnetization for ferromagnetic material is
 (a) zero (b) infinite
 (c) 10 (d) none of these (a)
28. At Neel temperature
 (a) susceptibility is maximum (b) susceptibility is minimum but not zero
 (c) susceptibility is zero (d) none of these (a)
29. The garnet crystal used in microprocessor has the formula
 (a) $Y_3Fe_5O_{12}$ (b) $Y_2Fe_3O_5$ (c) YFe_3O_5 (d) YFe_5O_{18} (b)
30. For paramagnetic materials, μ_r (relative permeability) at room temperature is
 (a) 10 (b) 5 (c) 1 (d) zero (c)
31. The susceptibility of a diamagnetic material is about
 (a) -10^{-5} (b) 10^{-3} (c) 10^5 (d) 10^7 (a)

SHORT QUESTION ANSWERS

1. How electric current results for most materials?
 Ans. For most materials, an electric current results from the motion of free electrons, which are accelerated in response to an applied electric field.
2. On what factor the number of free electrons in a material depend?

Ans. The number of free electrons in most materials depends on the electron energy band structure of the material. An electron energy band is just a series of electron states that are closely spaced with respect to energy, and one such band may exist for each electron subshell found in the isolate atom. By electron energy band structure is meant the manner in which the outermost bands are arranged relative to one another and then filled with electrons.

3. Can we consider the total energy of the free electrons in a solid where the potential field is uniform as their kinetic energy?

Ans. Yes. This energy varies as the square of the wave number k .

4. What is Fermi energy level?

Ans. This is defined as that level which has a 50% probability of occupation by an electron. At 0 K temperature, the Fermi energy level is the highest filled electron energy level.

5. What is the basic requirement for electrical conduction with respect to Fermi energy level in a solid?

Ans. The availability of empty electron states immediately above the Fermi energy level in a solid is a basic requirement for electrical conduction.

6. Where metals and alloys are chosen?

Ans. Metals are chosen where high conductivity and large temperature coefficient of resistance is desired. Alloys are chosen for heating purposes by I^2R dissipation and for low temperature coefficient of resistance.

7. What are the characteristics of dielectric materials?

Ans. Dielectric materials are electrically insulative, yet susceptible to polarization in the presence of an electric field. This polarization phenomenon accounts for the ability of the dielectrics to increase the charge storing capacity of capacitors, the efficiency of which is expressed in terms of a dielectric constant.

8. How polarization takes place in dielectrics?

Ans. Polarization results from the inducement by or orientation with the electric field of atomic or molecular dipoles; a dipole is said to exist when there is a net spatial separation of positively and negatively charged entities.

9. What are the possible polarization types in a dielectric?

Ans. These include ionic, electronic orientation and space charge. However, it is not necessary that all types be present in a particular dielectric. These processes have widely varying relaxation times.

10. Whether the dielectric constant of a material depends on the frequency of the alternating applied field and the temperature?

Ans. Yes.

11. When the energy loss for an dielectric in an alternating field is maximum?

Ans. When the relaxation time of a polarization process matches the period of the alternating field, the electrical energy loss of a dielectric is a maximum.

12. What will happen to the charged stored in a capacitor, if a dielectric material is inserted between the plates of a capacitor?

Ans. The charge stored in the capacitor increases due to polarization in the dielectric.

13. What are the main causes of electric breakdown of a dielectric?

Ans. The main causes of electric breakdown of a dielectric are impurities, surface flaws such as cracks and pores and absorption of moisture and gases on the surface.

14. What are ferroelectric materials?

Ans. These are the materials which may exhibit polarization spontaneously, i.e., in the absence of any external field and have very large dielectric constants. They are useful as piezoelectric crystals and as capacitors.

15. What is piezoelectricity?

Ans. This is the phenomenon whereby polarization is induced in a material by the imposition of external forces.

16. What is the cause of the macroscopic magnetic properties of a material?

Ans. These are as a consequence of interactions between an external magnetic field and magnetic dipole moments of the constituent atoms. Associated with each individual electron are both orbital and spin magnetic moments. The net magnetic moment for an atom is just the sum of the contributions of each of its electrons, where in there will be spin and orbital moment cancellation of electron pairs.

17. How diamagnetism results?

Ans. Diamagnetism results from changes in electron orbital motion that are induced by an external field. All materials are diamagnetic.

18. What are paramagnetic materials?

Ans. These are the materials having permanent atomic dipoles, which are acted on individually and are aligned in the direction of an external field.

19. Why dia and paramagnetic materials are considered to be non magnetic?

Ans. The magnetizations for these types of materials are relatively small and persist only while an applied field is present.

20. How, a large and permanent magnetization is established in ferromagnetic materials?

Ans. Due to bonding forces, the spins of electrons of neighbouring atoms are permanently aligned in ferromagnetic materials. The common ferromagnetic elements are Fe, Co and Ni.

21. What are antiferromagnetic materials?

Ans. Antiparallel coupling of adjacent cation spin moments is found for some ionic materials. Those in which there is total cancellation of spin moments are termed antiferromagnetic.

22. What are ferrimagnetic materials?

Ans. These materials have an antiparallel arrangement of electron spins, but due to incomplete cancellation, there is a net magnetic moment that is comparable in order of magnitude to that of ferromagnetic materials.

23. How the net magnetization results for cubic ferrites?

Ans. The net magnetization for cubic ferrites results from the divalent ions (e.g., Fe^{2+}) that reside on octahedral lattice sites, the spin moments of which are mutually aligned.

24. How ferro and ferrimagnetic materials behave with temperature?

Ans. With rising temperature, increased thermal vibrations tend to counteract the dipole coupling forces in ferro- and ferrimagnetic materials. Consequently, the saturation magnetization gradually diminishes with temperature upto the curie temperature (T_c), at which point it drops to zero; above T_c , these materials are paramagnetic.

25. Explain domain character of ferro and ferrimagnetic materials.

Ans. Below curie temperature (T_c), a ferro-or ferrimagnetic material is composed of domains-small volume regions where in all net dipole moments are mutually aligned and the magnetization is saturated. The total magnetization of the material is just the appropriately weighted vector sum of the magnetizations of all these domains. When an external magnetic field is applied, domains having magnetization vectors oriented in the direction of the magnetic field grow at the expense of domains that have unfavourable magnetization orientations. At total saturation the entire solid material is a single domain and the magnetization is aligned with the field direction. The change in domain structure with increase or reversal of a magnetic field is accomplished by the motion of domain walls.

26. How will you explain both hysteresis and permanent magnetization in terms of domain walls?

Ans. Hysteresis (the lag of B field behind the applied H field) and permanent magnetization (or remanence) result from the resistance to movement of domain walls.

27. What are soft magnetic materials?

Ans. Domain wall movement in these materials is easy during magnetization and demagnetization. Consequently they have small hysteresis loops and low energy losses, i.e. low eddy current losses. We may note that by increasing the electrical resistivity of the magnetic medium eddy current losses may be reduced.

28. What are hard magnetic materials?

Ans. Domain wall motion in these materials is much more difficult, which results in larger hysteresis loops; because greater fields are required to demagnetize these materials, the magnetization is more permanent.